

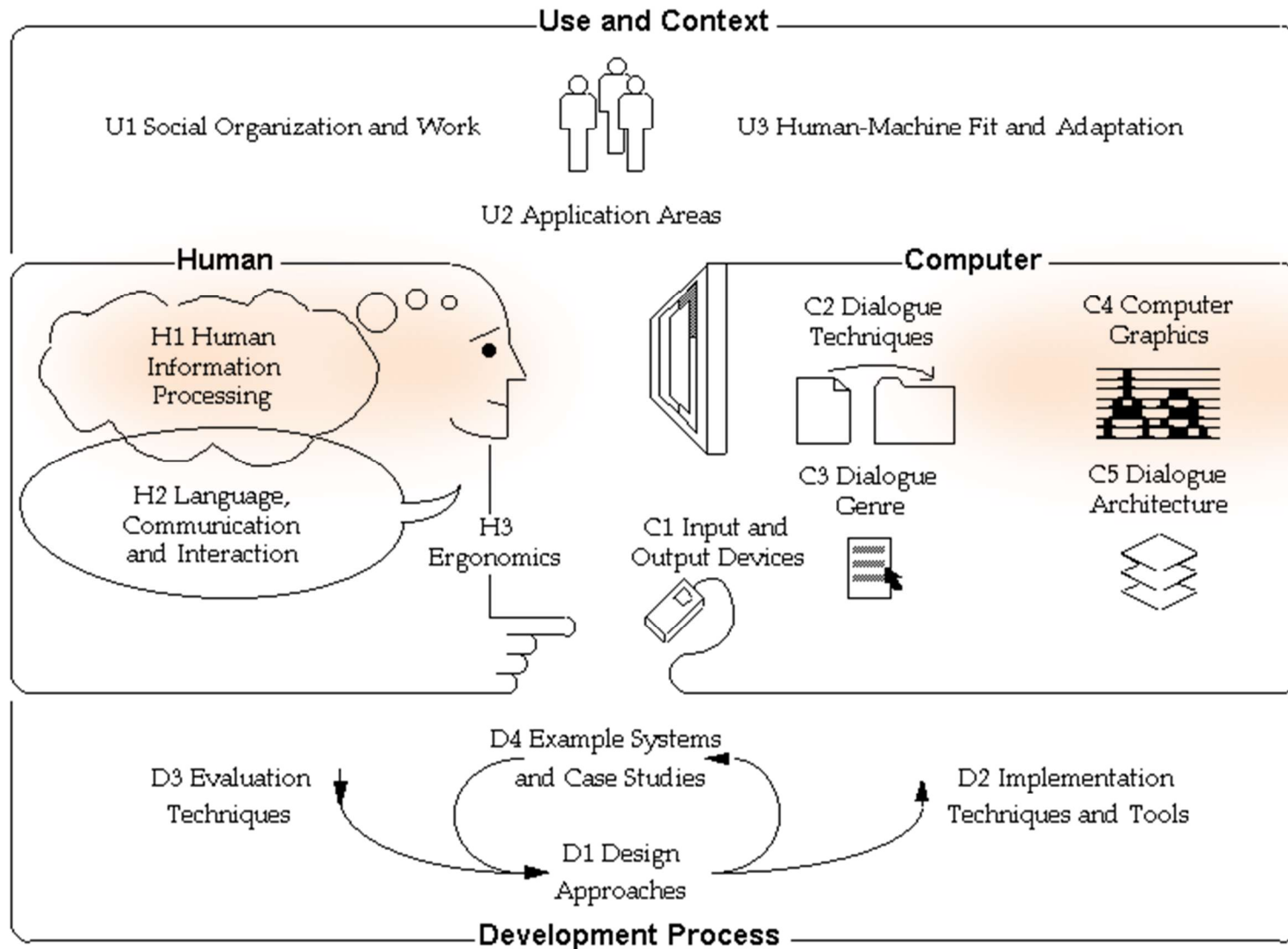
HCI Methods

Johan F. HOORN and Jeff TANG

Lecture 03, Part 2: Visual perception and computer graphics



Overview: Remember where we were



Reading text

(= visual pattern recognition)

Input: perception

❑ Visual

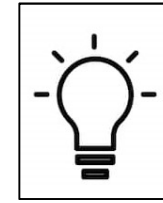
- Text is an image with coded meaning

Three stages of reading:

- Visual pattern perceived
 - Decoded using internal representation of language
 - Interpreted using knowledge of syntax, semantics, and pragmatics
- Icons aren't always the most appropriate
 - Sometimes textual descriptions work better



What do
these
icons
mean?



This
is
text

Debug.Print text

Input: perception

□ Visual

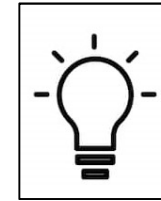
- Text is an image with coded meaning

Three stages of reading:

- Visual pattern perceived
 - Decoded using internal representation of language
 - Interpreted using knowledge of syntax, semantics, and pragmatics
- Icons aren't always the most appropriate
 - Sometimes textual descriptions work better



Bar
Restaurant
Self Service
Snacks



This
is
text

Debug.Print text

Input: perception

□ Visual

- Text
 - Decoding and interpretation

Syntactic

Language structure, grammar, for instance:

Sentence = Noun_Phrase Verb_Phrase or Debug.Print text

Semantic

The (non)literal meanings of words and sentences, regardless of context, for instance:

“Sihk jó faahn meih a?” (Have you eaten yet?)

Pragmatic

Meaning with regard to the individuals involved and the context of the situation

→ “How do you do?” or “Sit and eat”

Input: perception

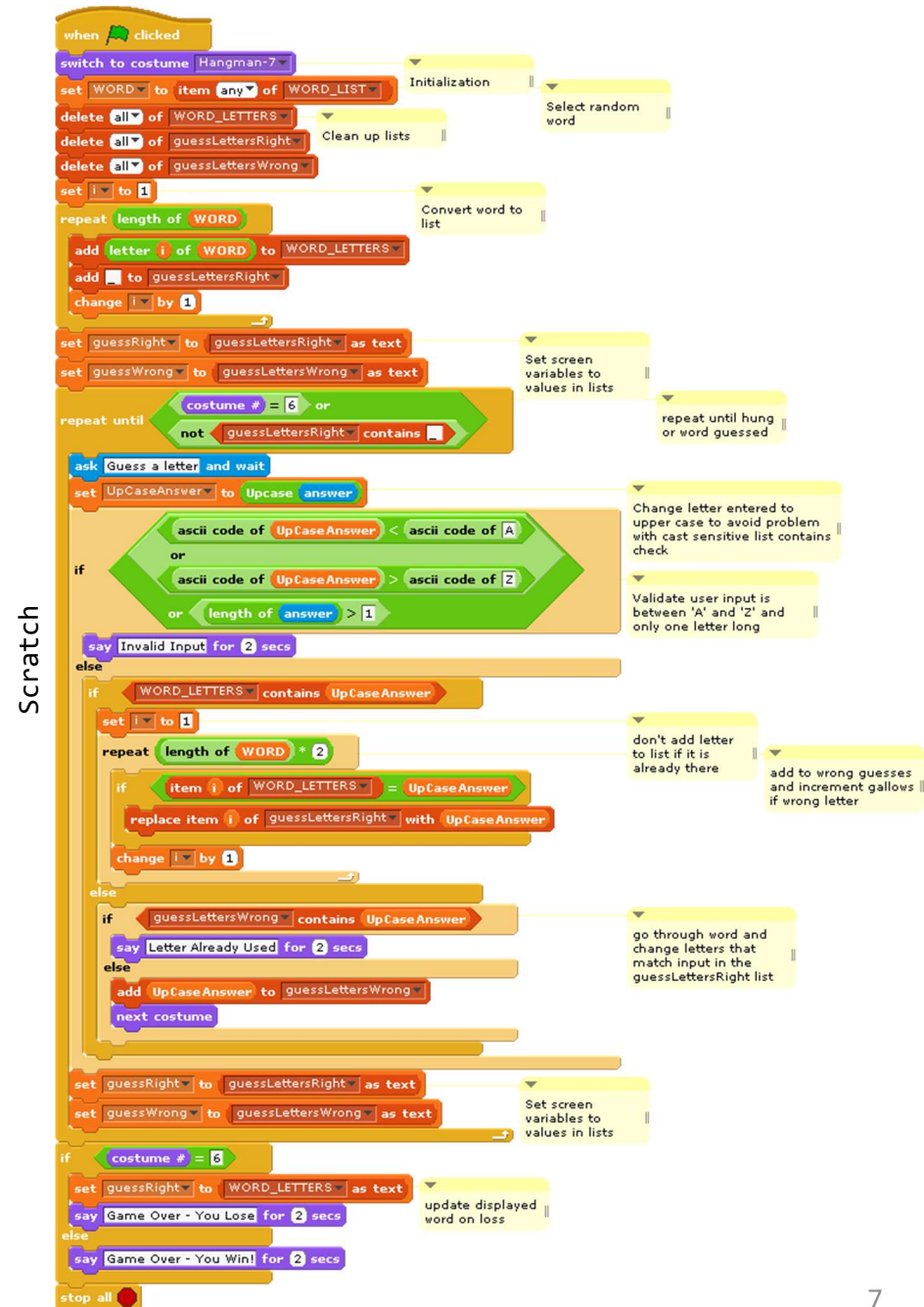
□ Visual

- Text
 - Decoding and interpretation

Semantics
change due to
Pragmatics

With time, comments are copied and the underlying code is altered, changing its original meaning

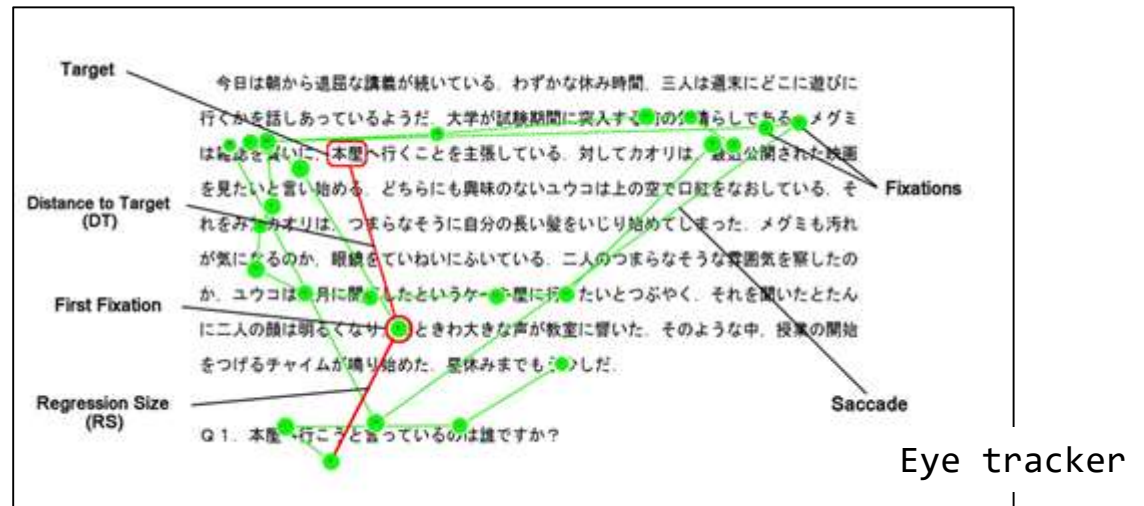
(Example by Al Gifford,
Feb. 10, 2013)



Input: perception

□ Visual

- Text
 - Reading



Scanning

Global exploration of the text and its structure, effort estimate (much/little text, long/short words)

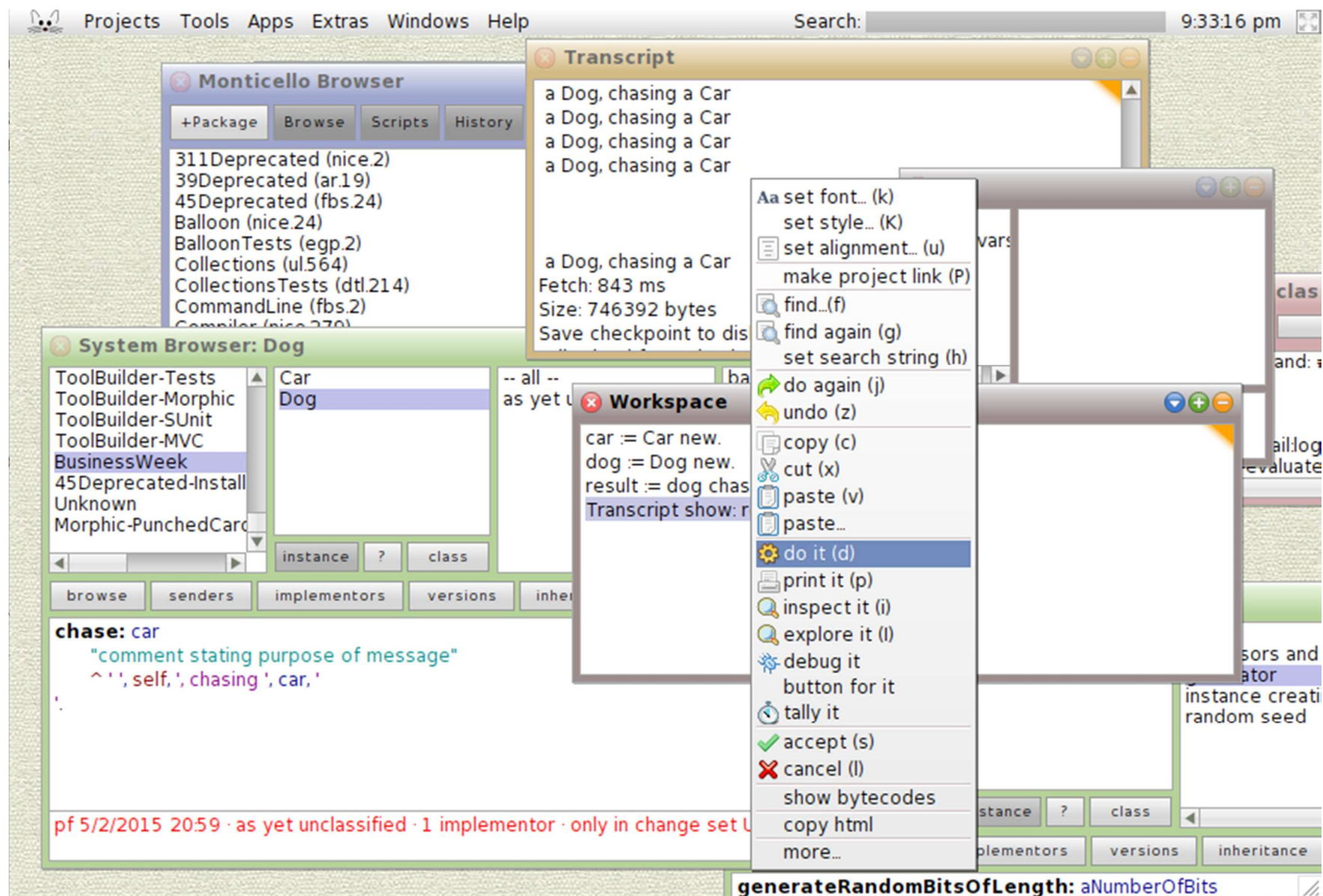
Saccades

Eye jumps, not just into reading direction but back and forth (and some up and down). Word shape (and subsequently, fonts) important for recognition. Top part of font more so than bottom

Fixations

Perception and comprehension takes place only during fixations (in between is idle state)

Scanning (eh, bit busy?)



Saccades



TheWindowsClub

News Windows Downloads Security Office Phone General Reviews Forum About

Web fonts appear blurry in Internet Explorer

RECOMMENDED: [Click here to repair Windows problems](#)

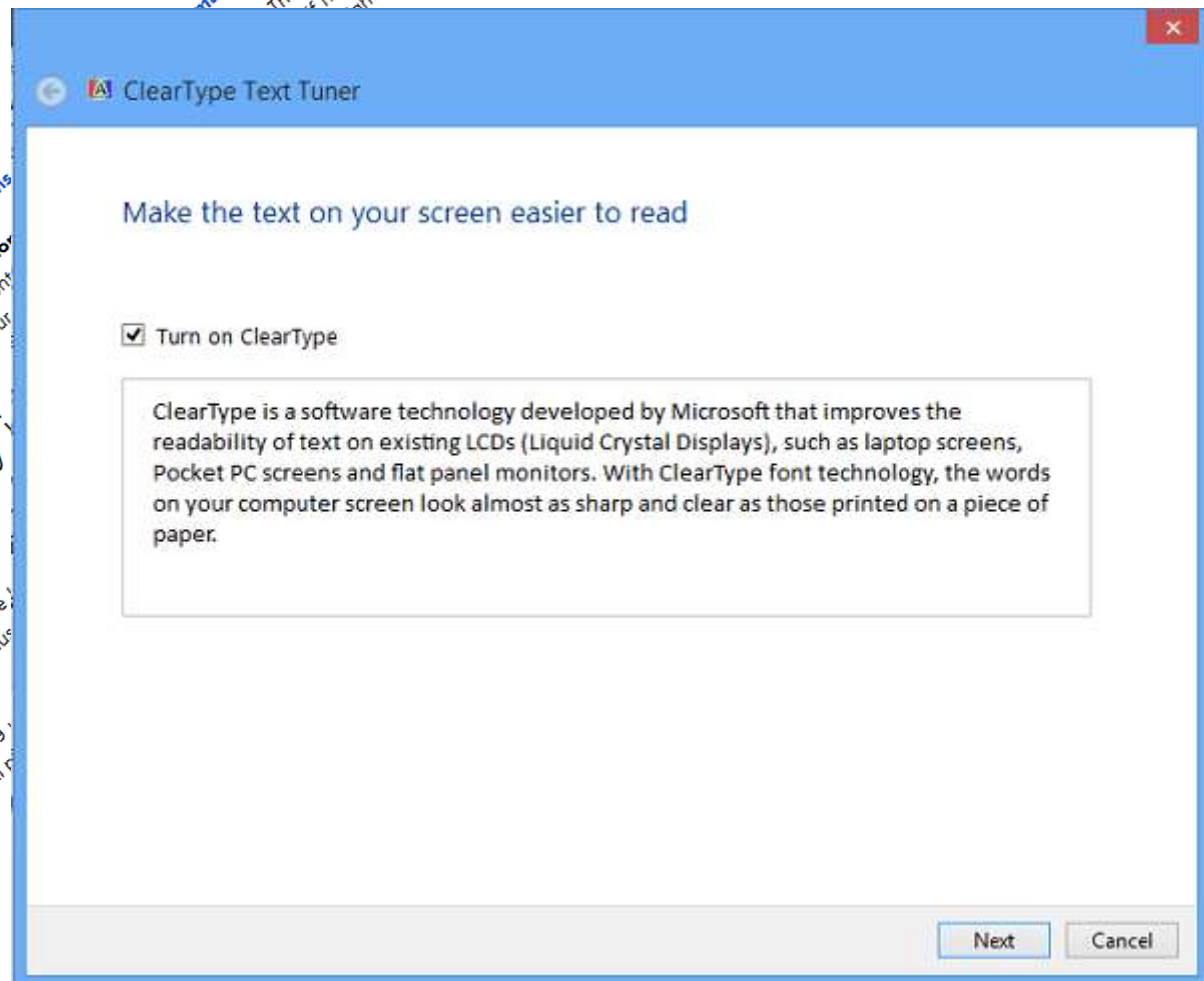
You may have observed, while viewing webpages in Internet Explorer, that the font quality may not persist in other versions of Internet Explorer i.e., the font quality you can improve the quality of fonts as they appear on your screen.

Web fonts appear blurry in Internet Explorer

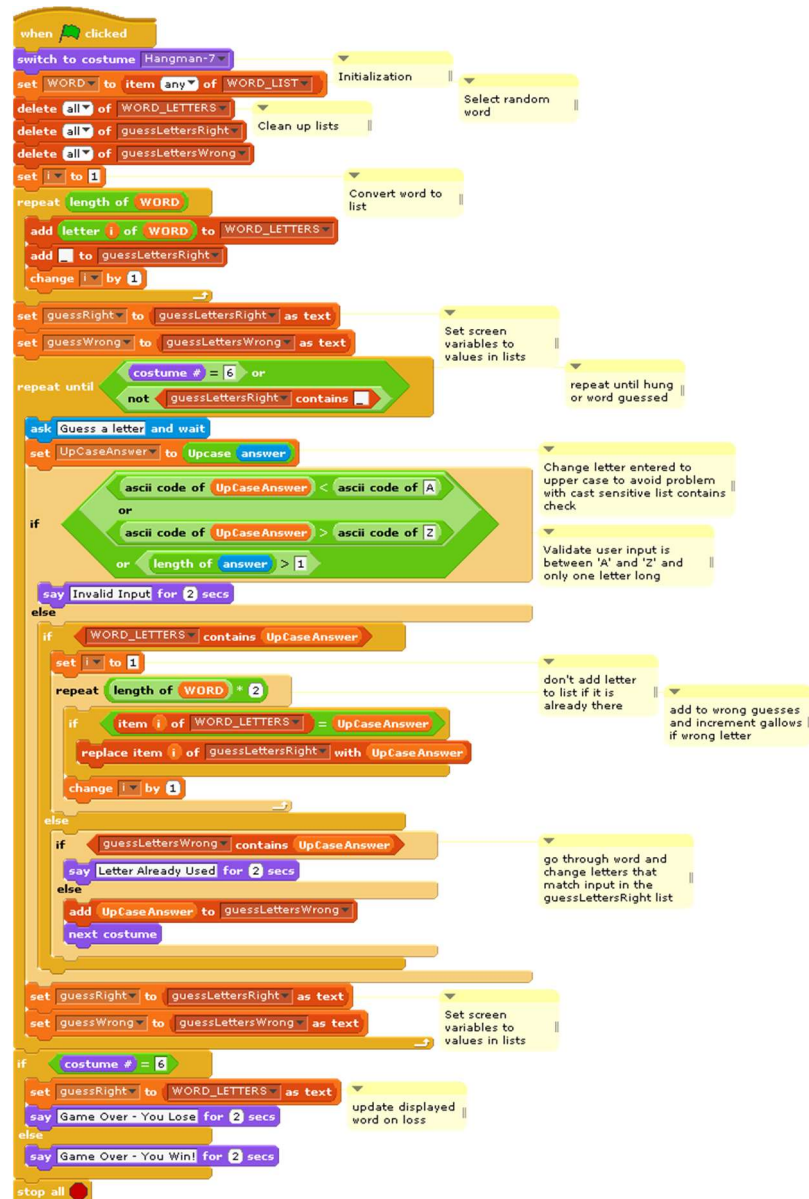
Update Fonts

You can update some fonts to improve the font quality in Internet Explorer 9. The problem occurs because of the way that Internet Explorer 9 renders text.

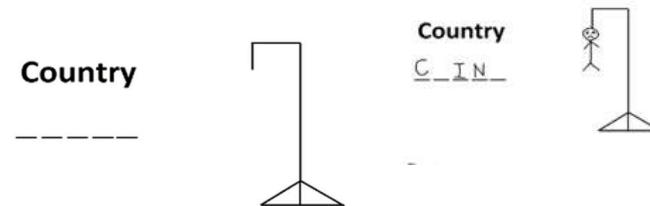
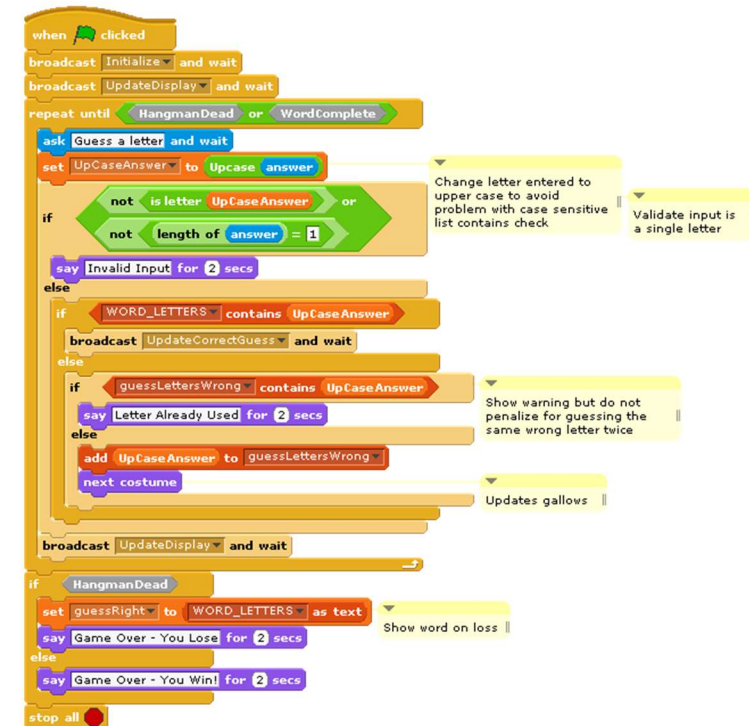
By default, Internet Explorer 9 uses subpixel rendering (GDI). Internet Explorer 8 uses whole-pixel rendering (GDI).



Fixations



Reduce number of fixations by writing code concisely



Input: perception

□ Visual

- Text

- Decoding and interpretation in design

Lexical ambiguity in interfaces

Semantic priming

The use of context

```
func loadFacebookUserDetails(){  
    // Show indicator  
    let spinningActivity = MBProgressHUD.showHUDAddedTo(self.view, animated: true)  
    spinningActivity.labelText = "Loading"  
    spinningActivity.detailsLabelText = "Please wait."  
  
    // Define fields we would like to read from Facebook User object  
    let requestParameters = ["fields": "id, email, first_name, last_name, name"]  
  
    // me  
    let userDetails = FBSDKGraphRequest(graphPath: "me", parameters: requestParameters)  
    userDetails.startWithCompletionHandler ( { (connection, result, error: NSError!) -> Void in  
        if error != nil{  
            Utility().displayMyAlertWithCustomTitleMessageAndButtonMessage("Error", userMessage: error!.  
                localizedDescription, buttonMessage: "OK", userVI: self)  
            PFUser.logout()  
            return  
        }  
  
        // Extract user fields  
        let userId: String = result["id"] as! String  
        let userEmail: String? = result["email"] as? String  
        //let userFirstName: String? = result["first_name"] as? String  
        //let userLastName: String? = result["last_name"] as? String
```

! Ambiguous use of 'subscript'

“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94

POST

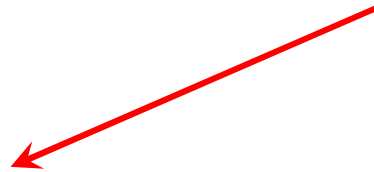
“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94



POST

Mail?



“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94



POST

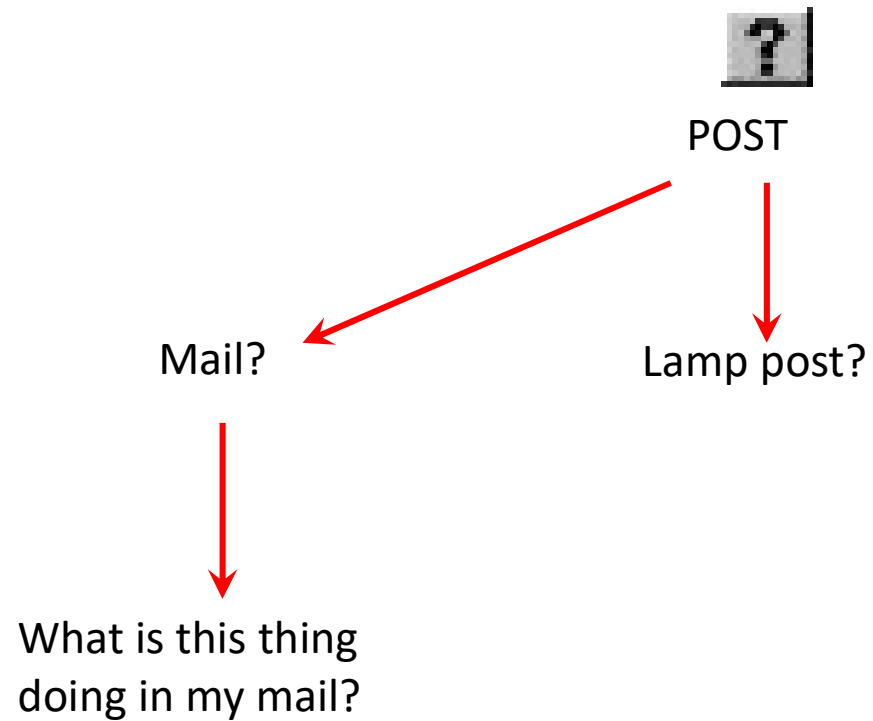
Mail?



What is this thing
doing in my mail?

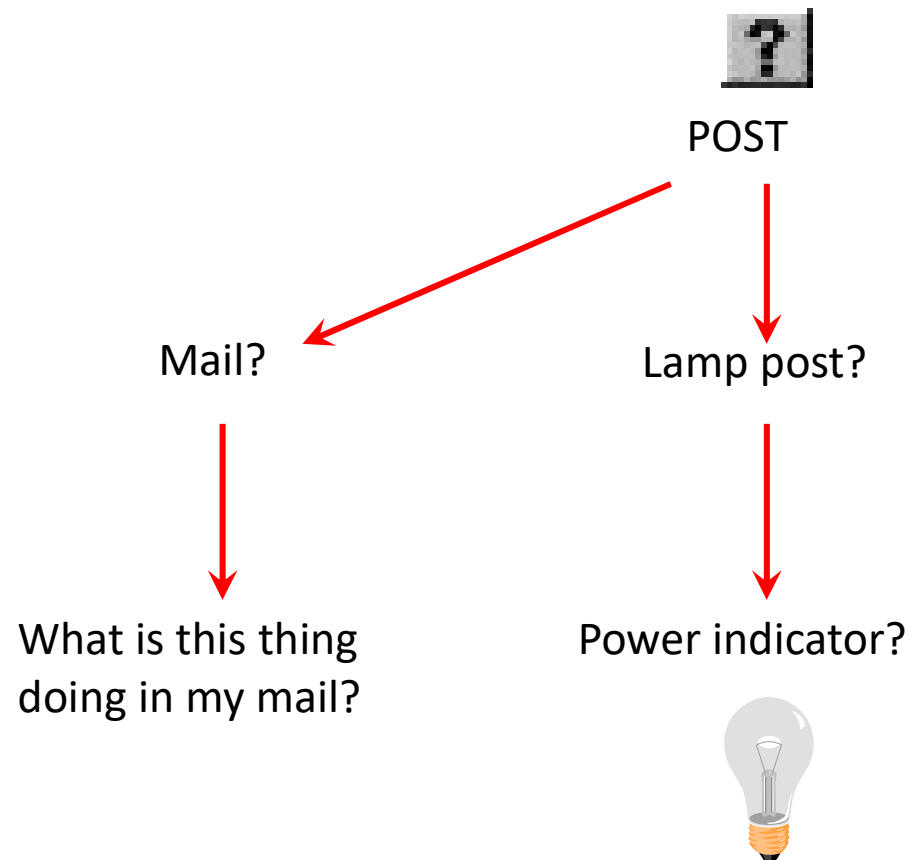
“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94



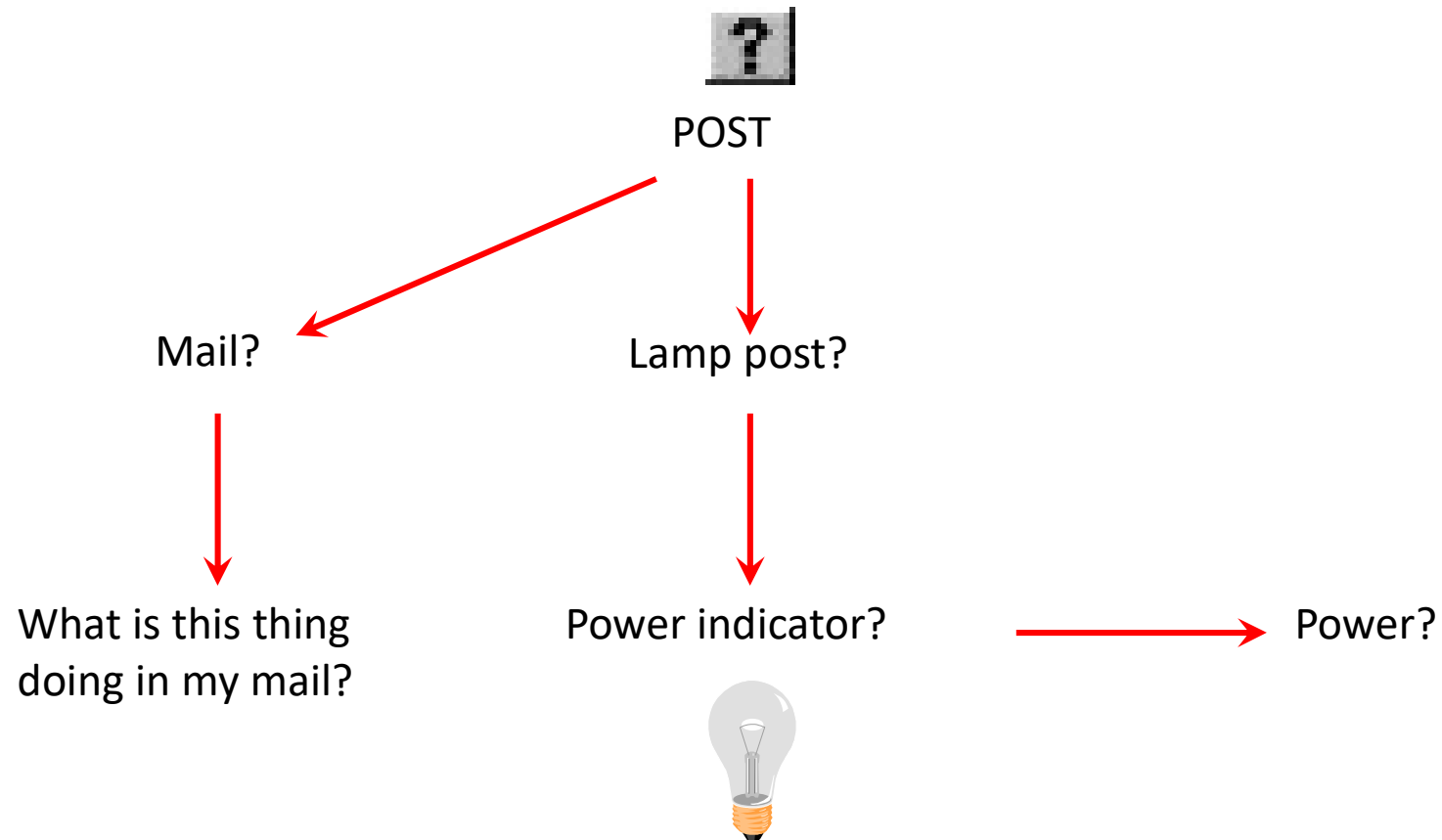
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ASUS (2001). *Notebook PC User's Manual*, p. 94



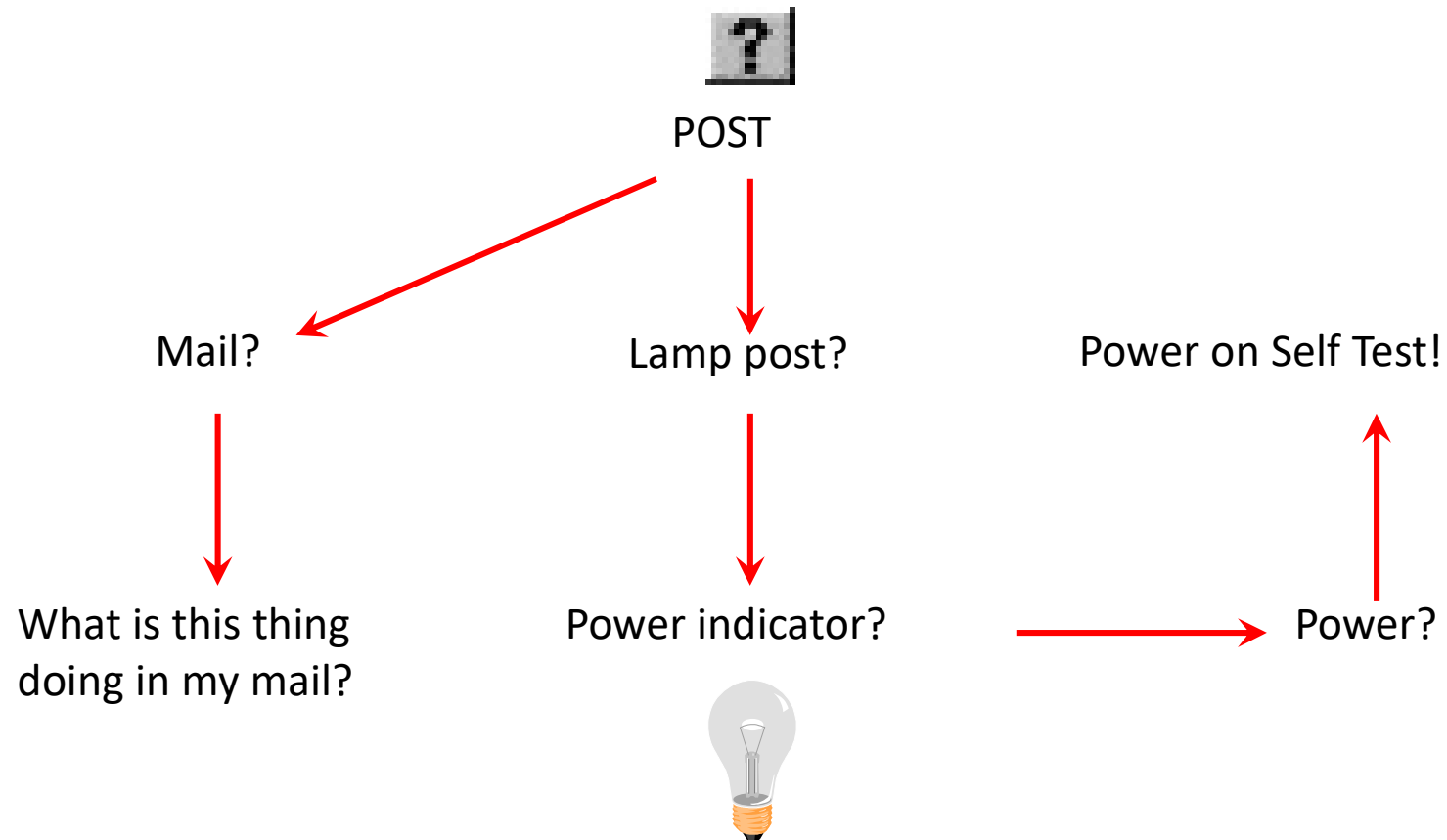
“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94



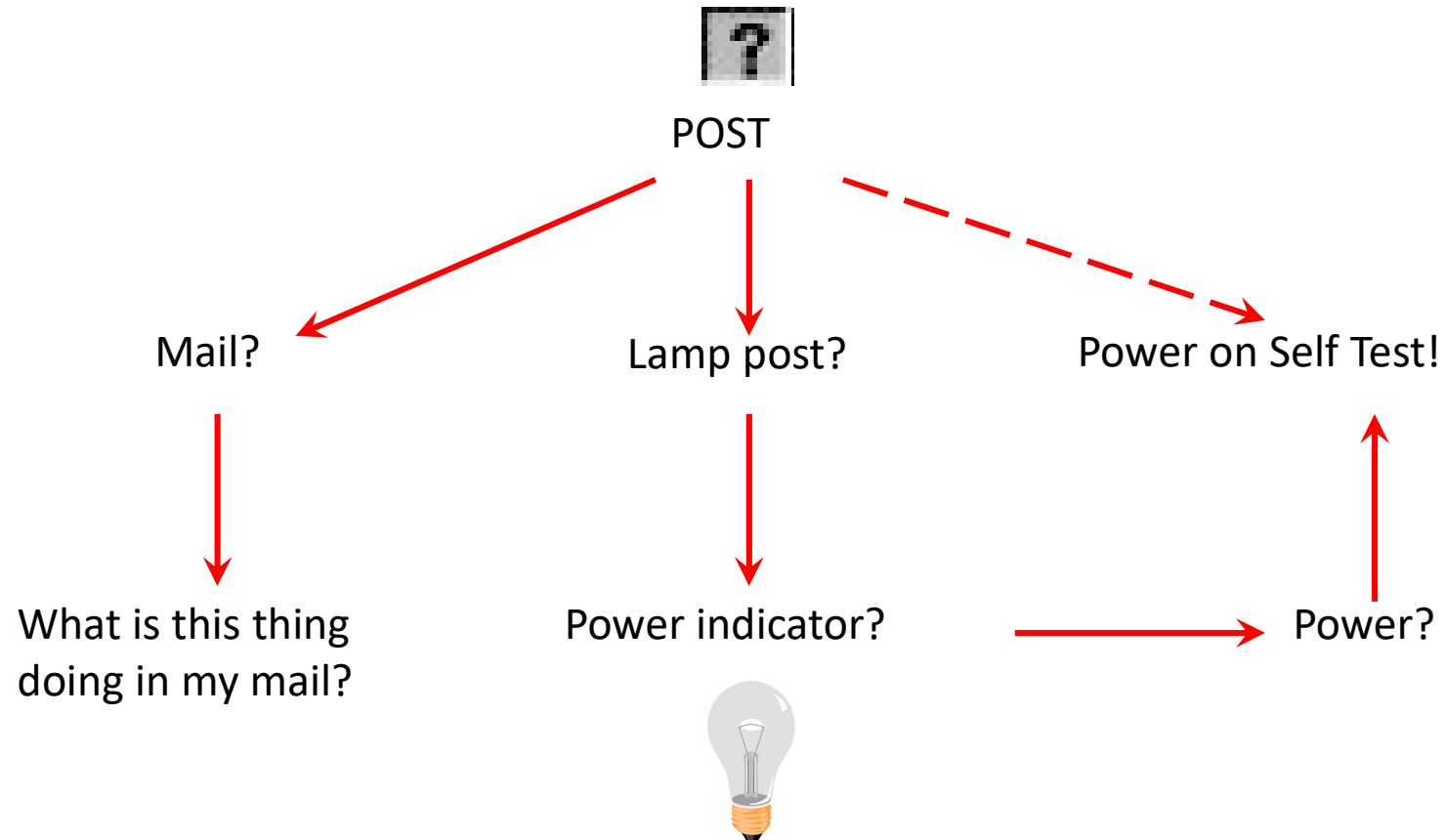
“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94



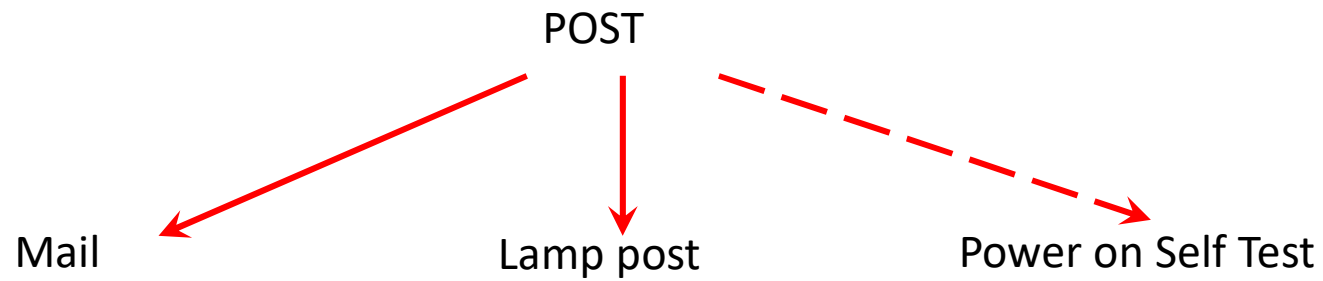
“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94



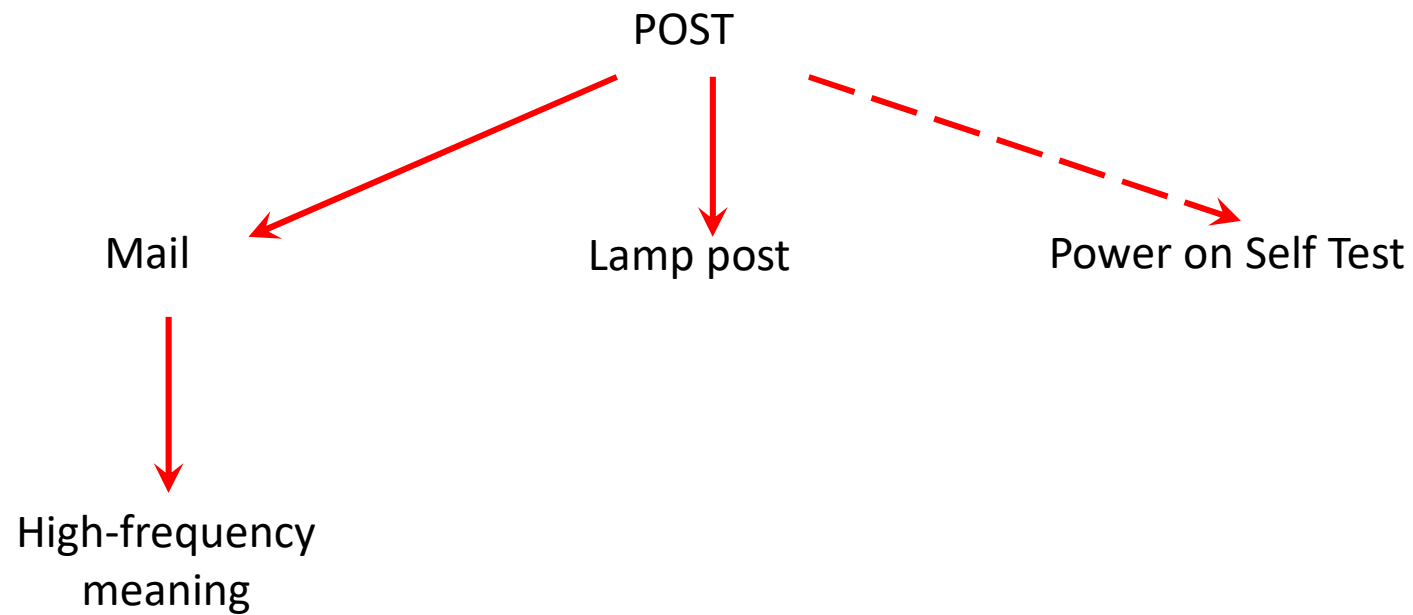
“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94



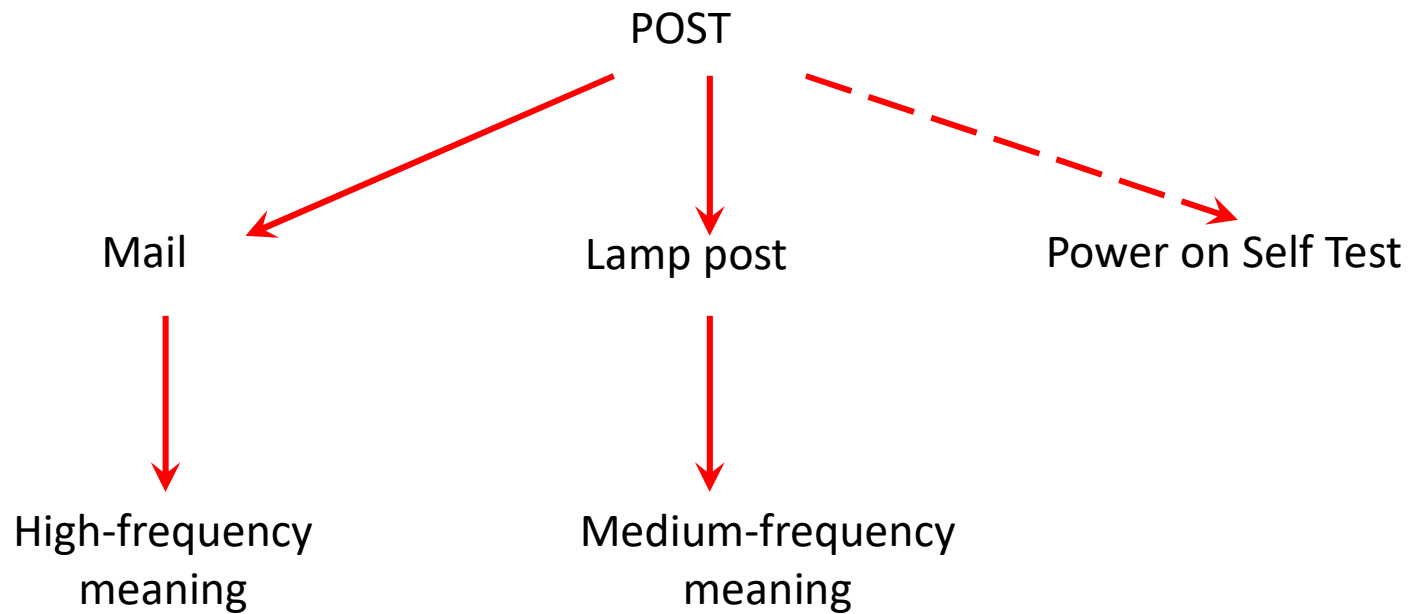
“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94



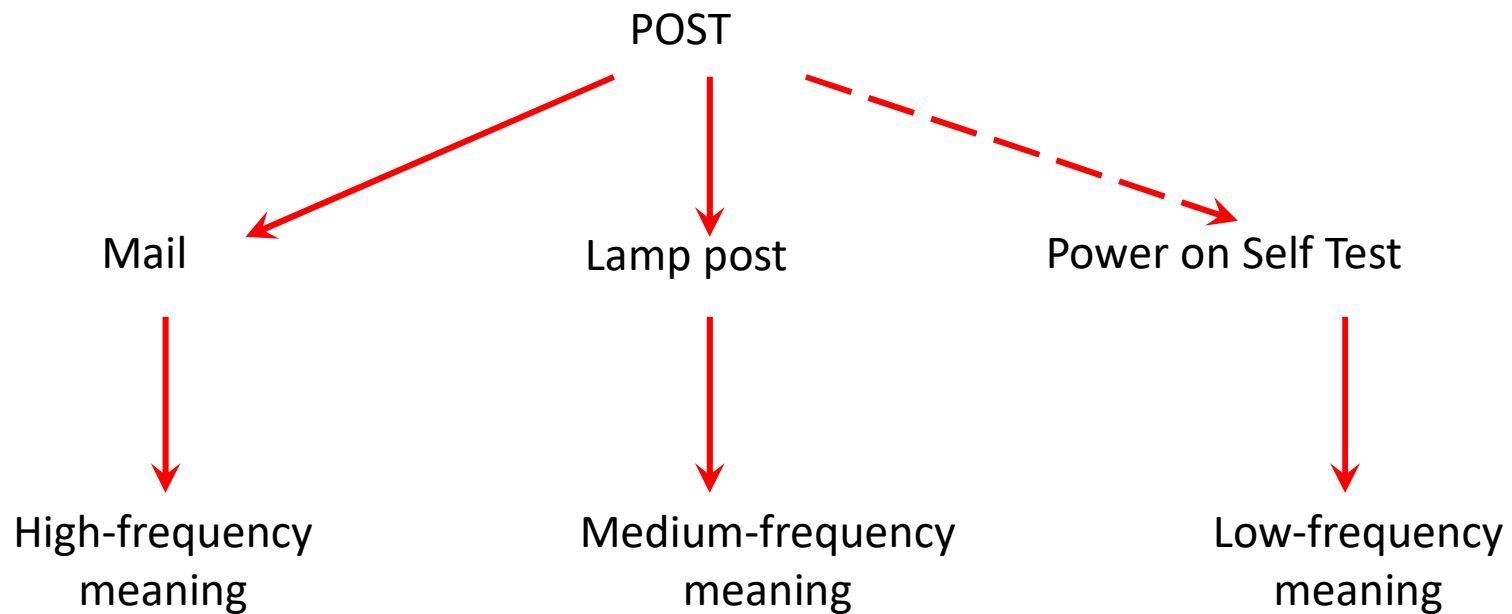
“When you turn on your computer,
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“When you turn on your computer,
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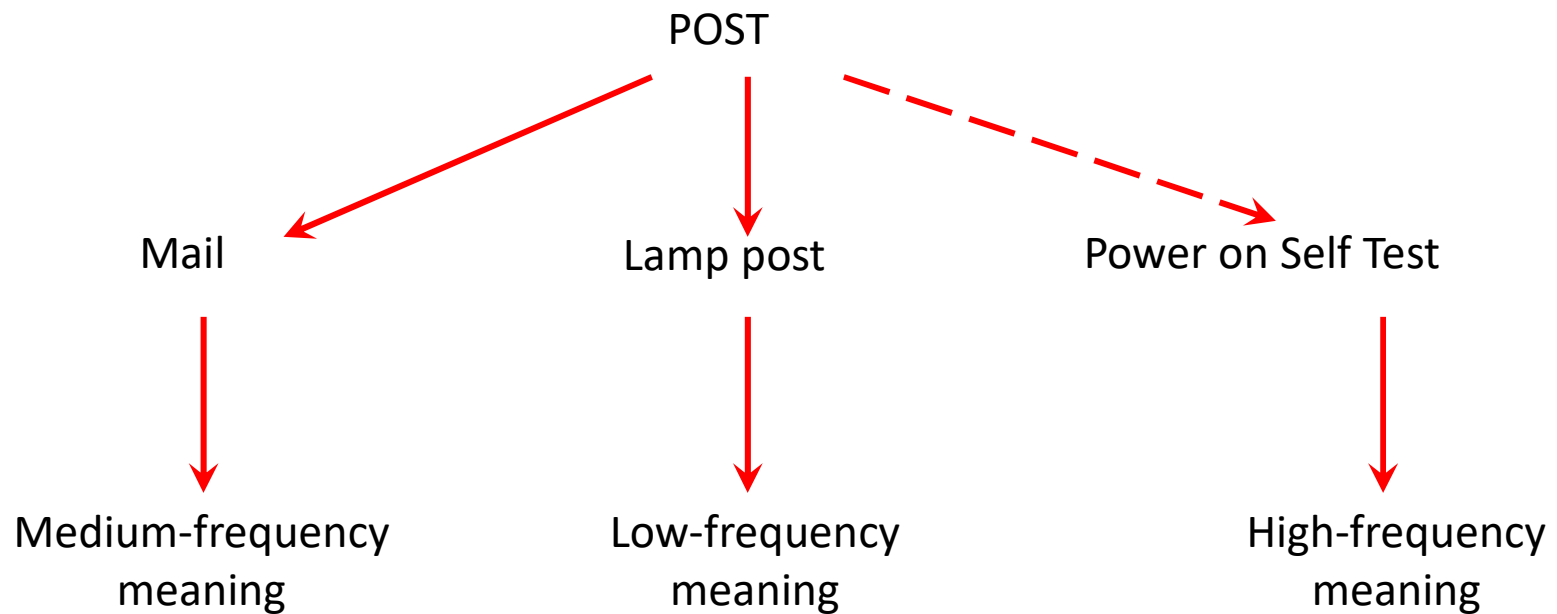
ASUS (2001). *Notebook PC User's Manual*, p. 94



In group of non-expert users!

“When you turn on your computer,
it will first run through the POST (...)”

ASUS (2001). *Notebook PC User's Manual*, p. 94



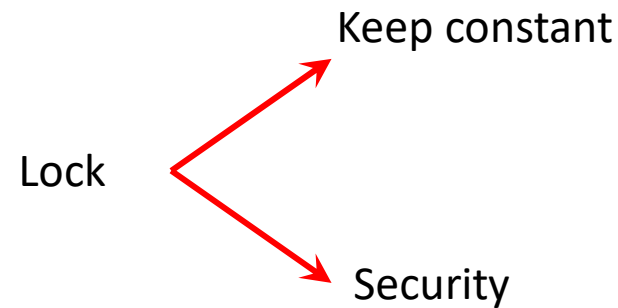
In group of domain-expert users!

The less you know as a user (novice), the more you rely on context.

If no context is provided, the dictionary meanings become active.

Like computers, users become error-prone with ambiguities.

Semantic priming is the solution.



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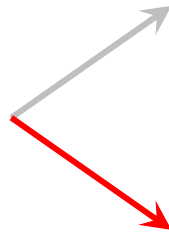
Semantic priming is the solution.

Prime:

Cable



Lock



Keep constant

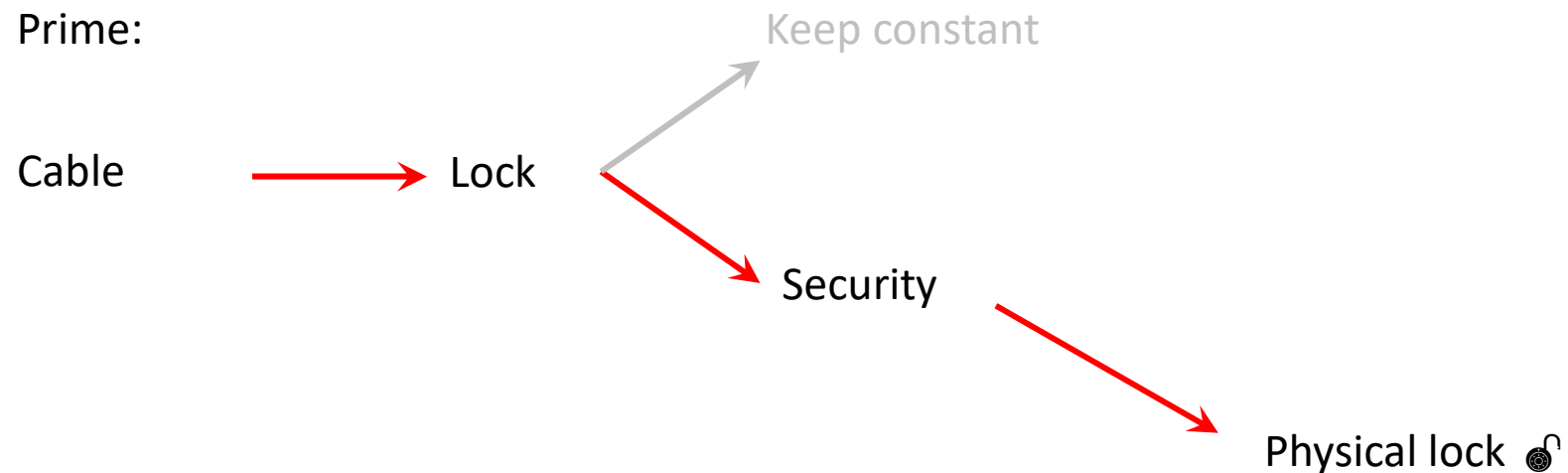
Security

The less you know as a user (novice), the more you rely on context.

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Semantic priming is the solution.



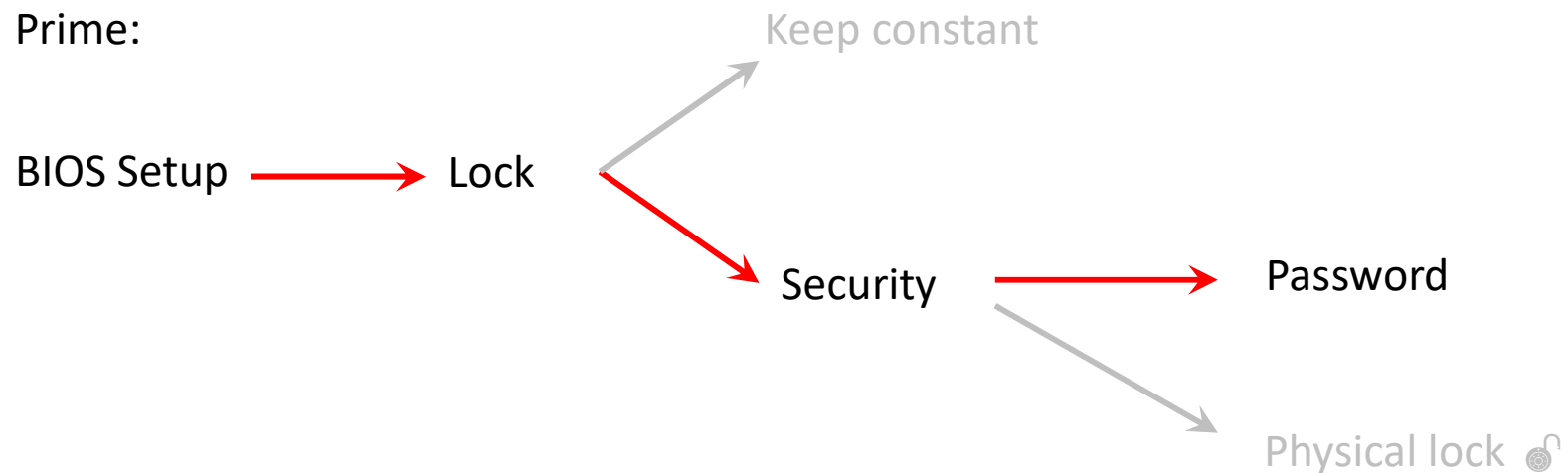
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Prime:

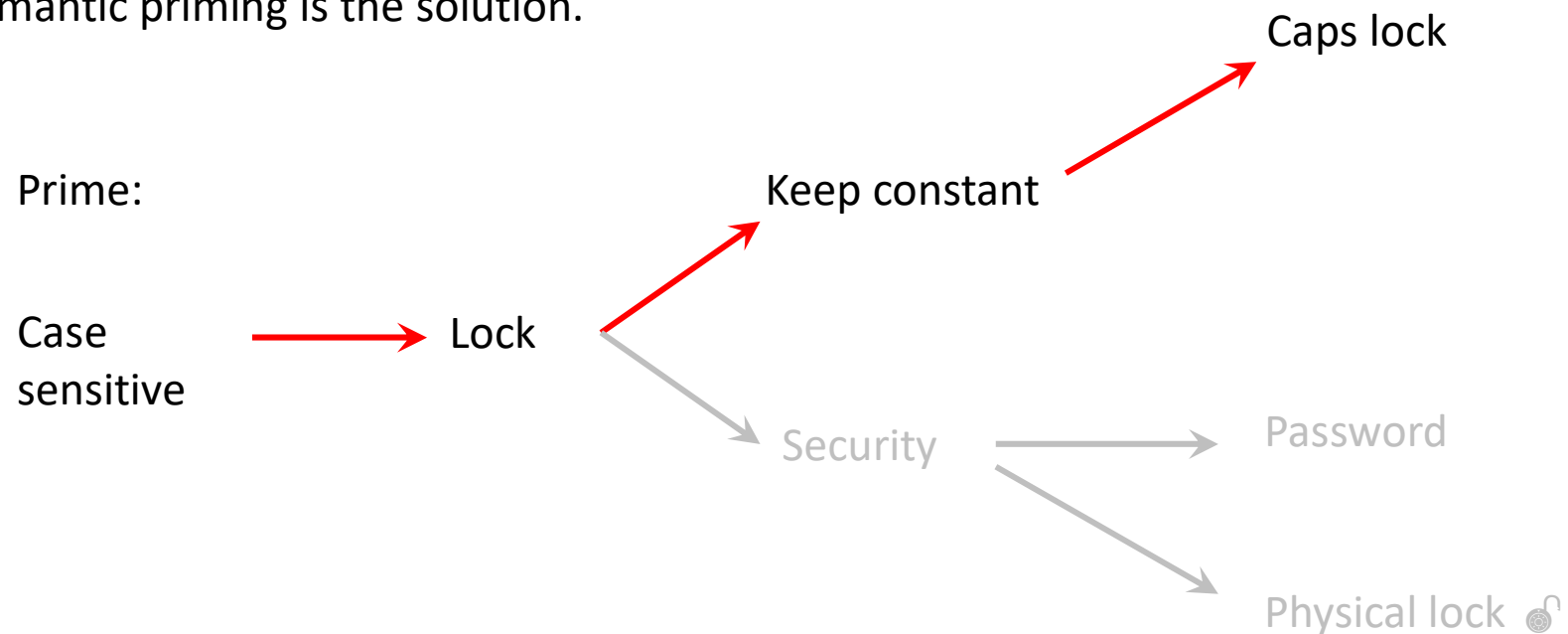


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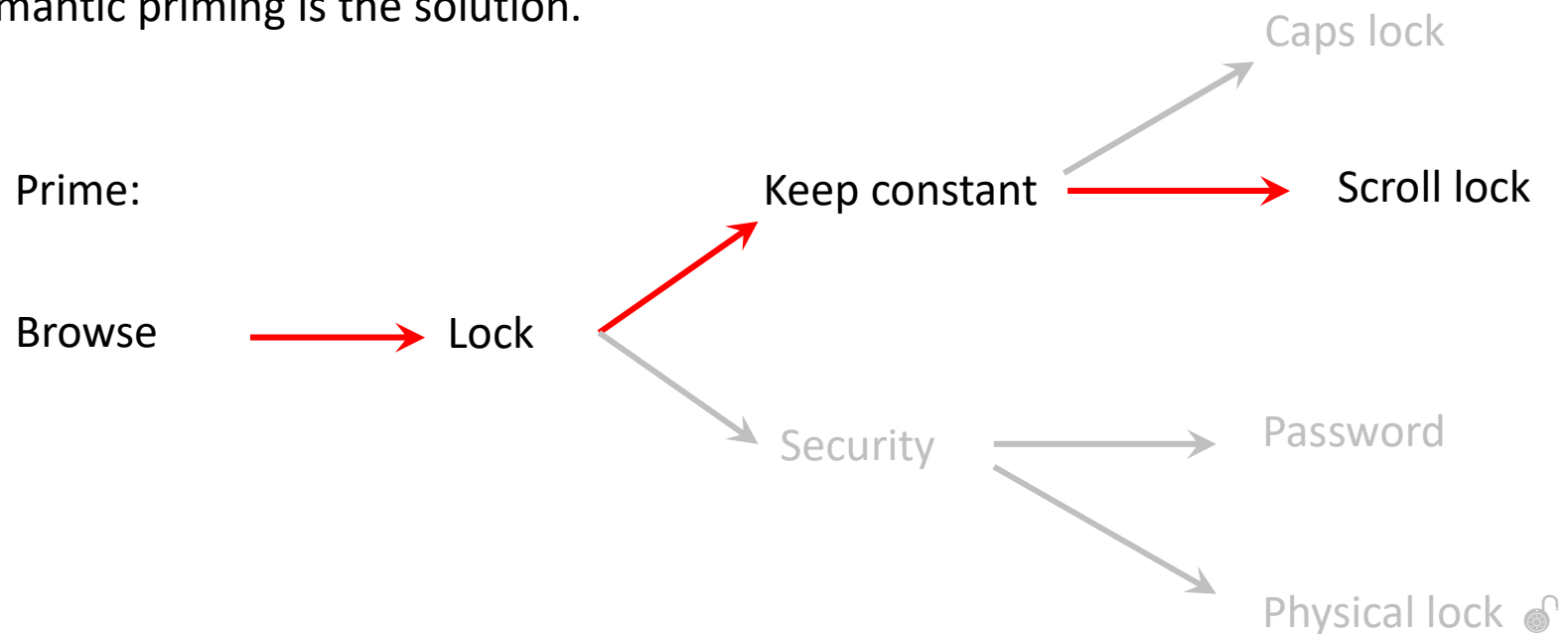


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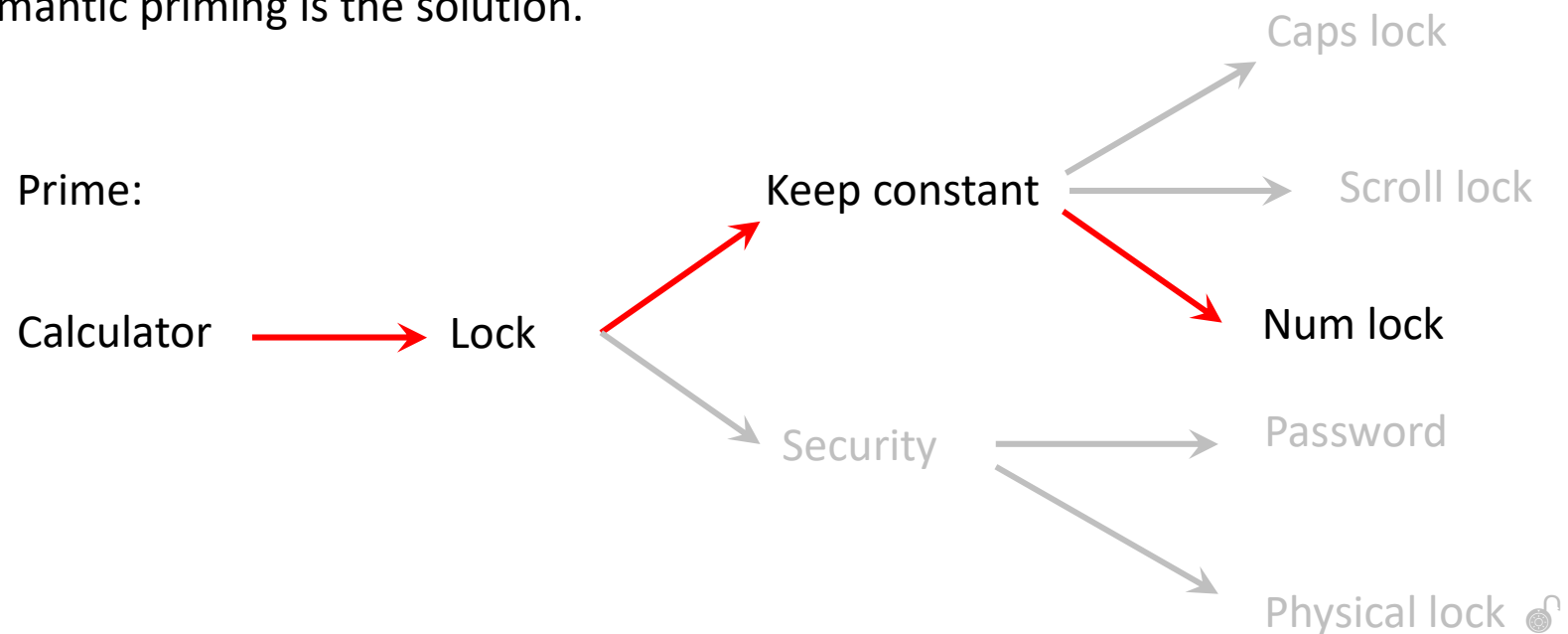


The less you know as a user (novice), the more you rely on context.

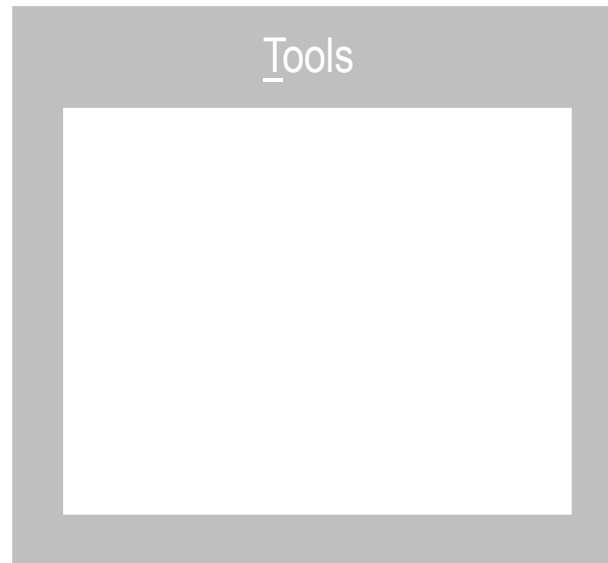
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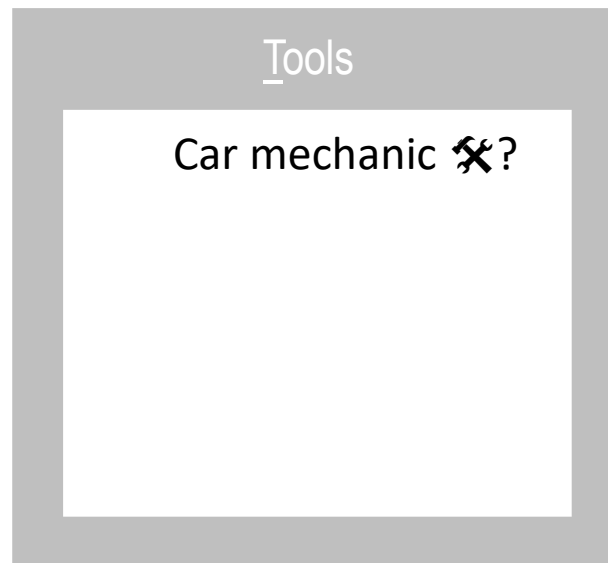
Semantic priming is the solution.



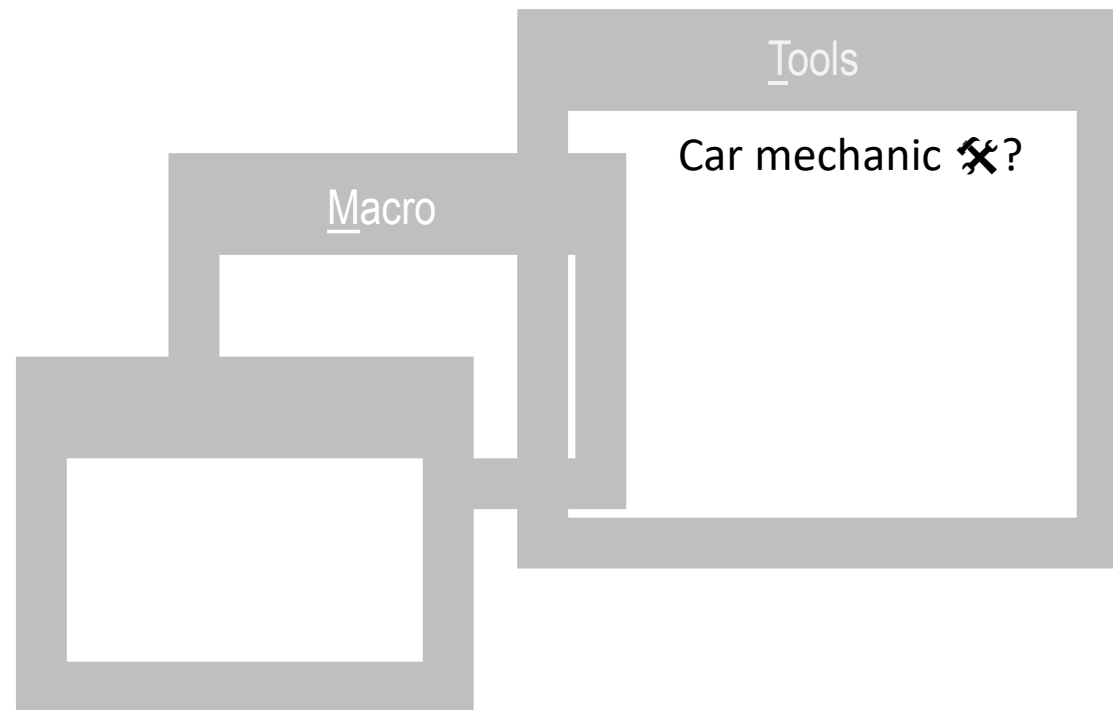
The same is valid for (linear) menu systems.



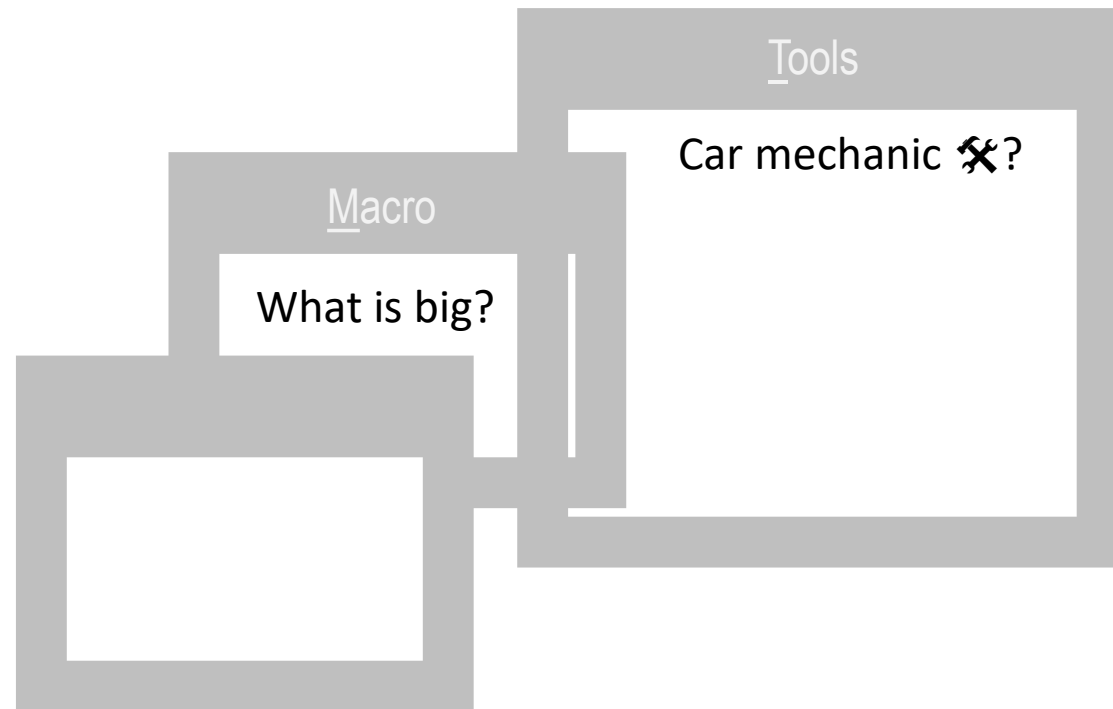
The same is valid for (linear) menu systems.



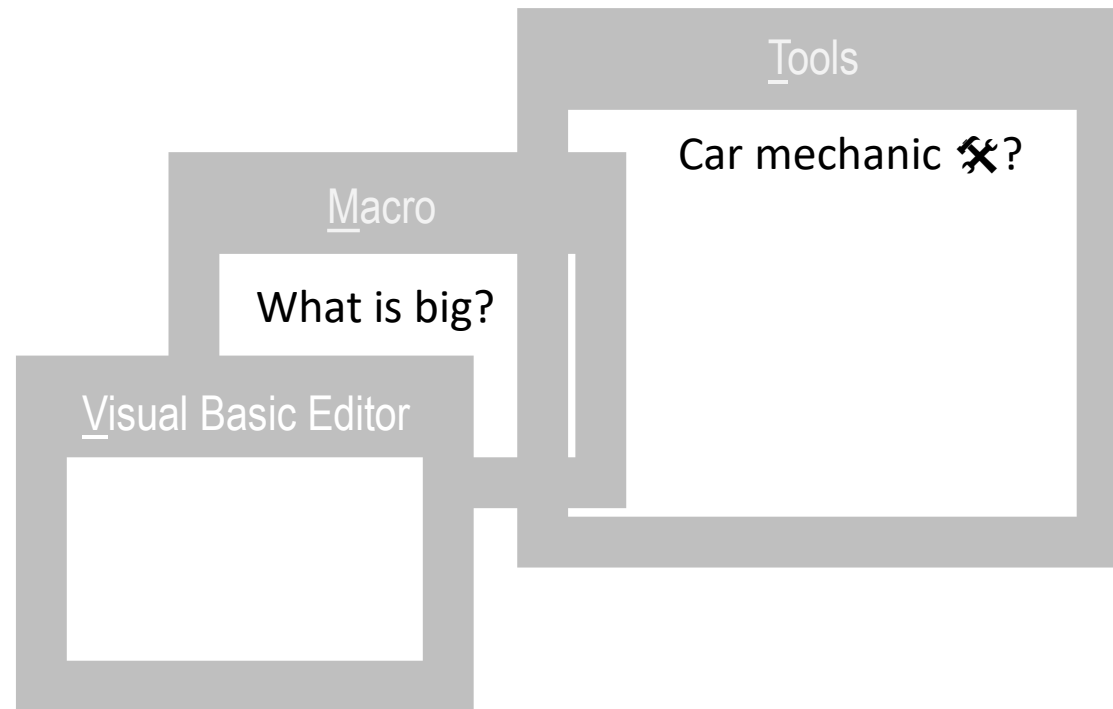
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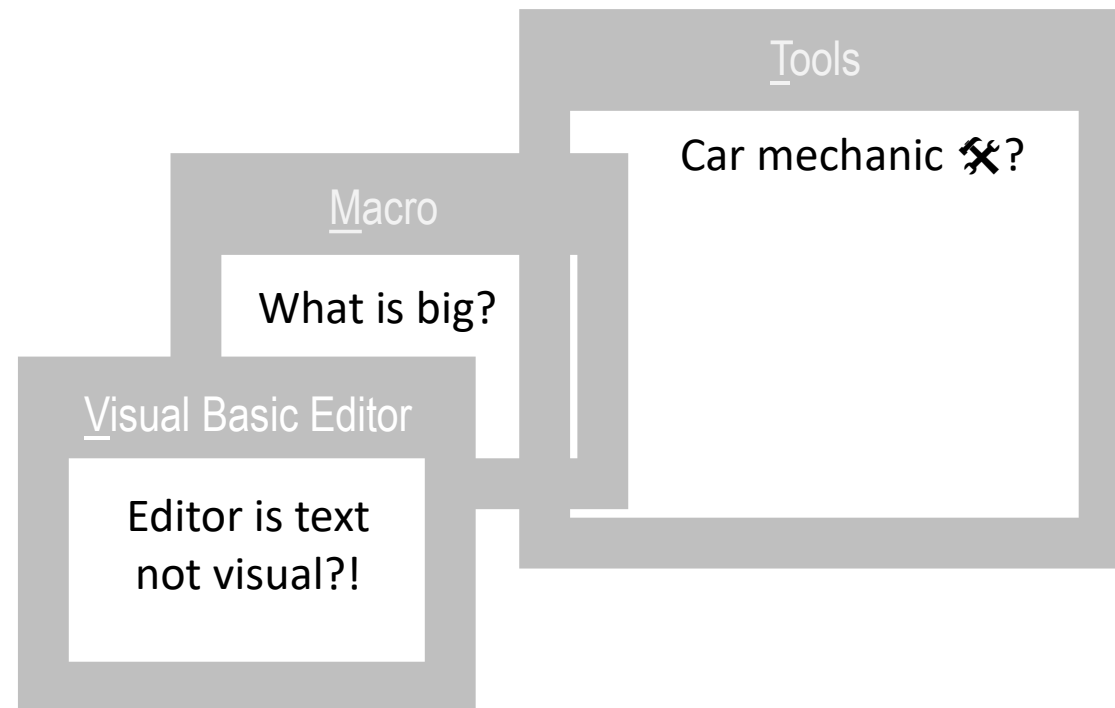
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The same is valid for (linear) menu systems.

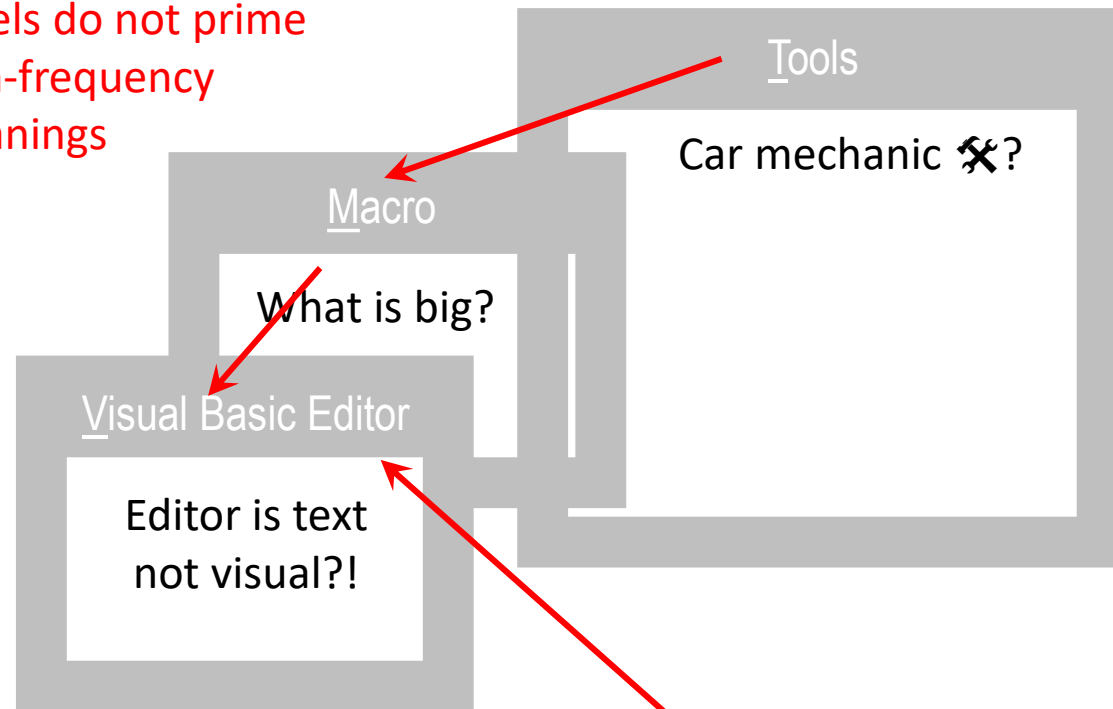


The same is valid for (linear) menu systems.



The same is valid for (linear) menu systems.

Labels do not prime
high-frequency
meanings

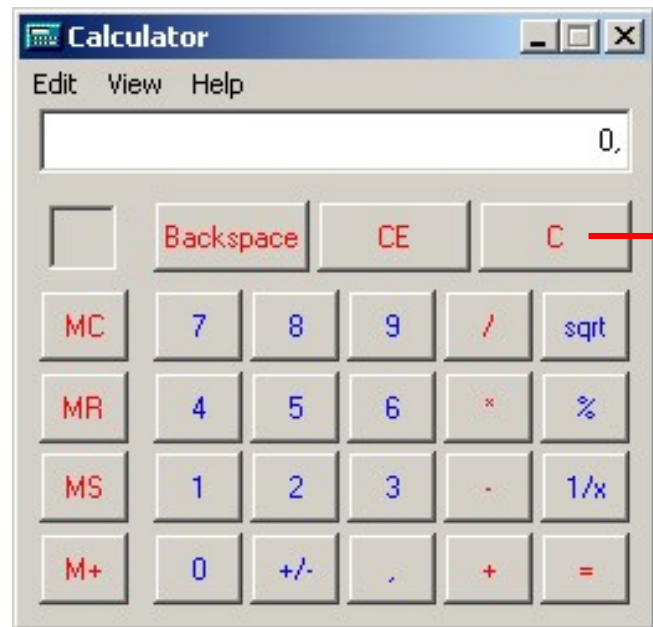


Prime does not resolve
ambiguity of name

What does this C mean?

C

What does this C mean?



C

Clear



What does this C mean?

Original thought of the designers

C

Clear



What does this C mean?

Actual thought of the end-users

C

Clear

Call

So, what does this C mean?

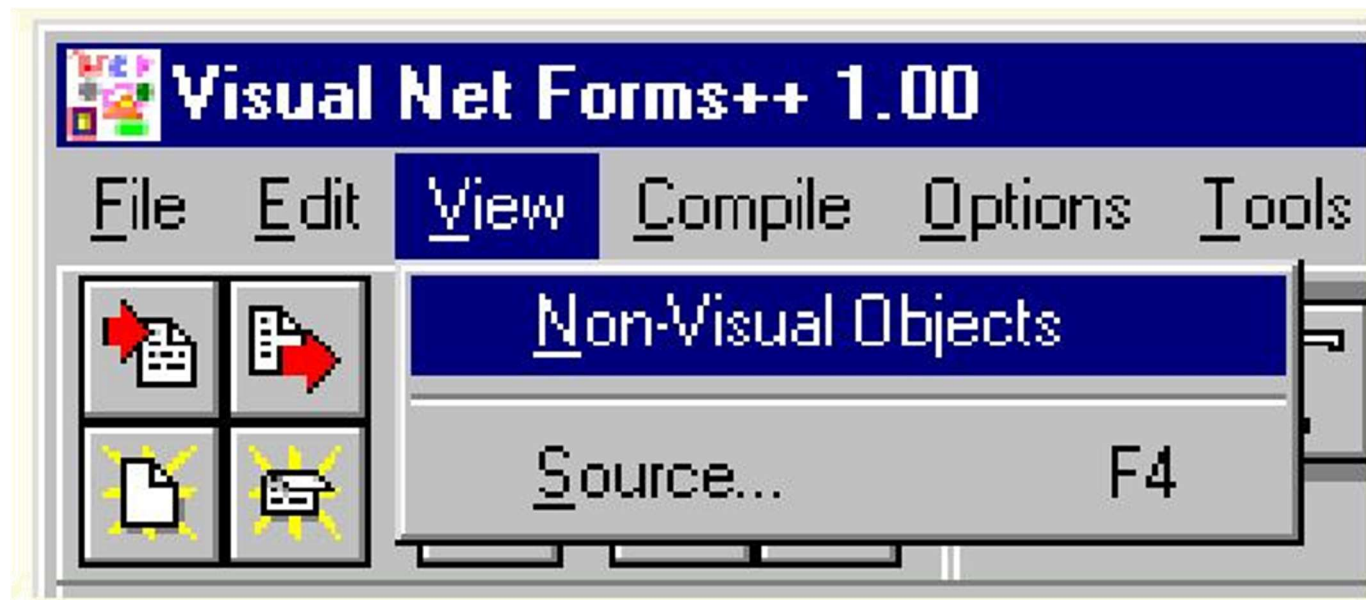
C

So, what does this C mean?

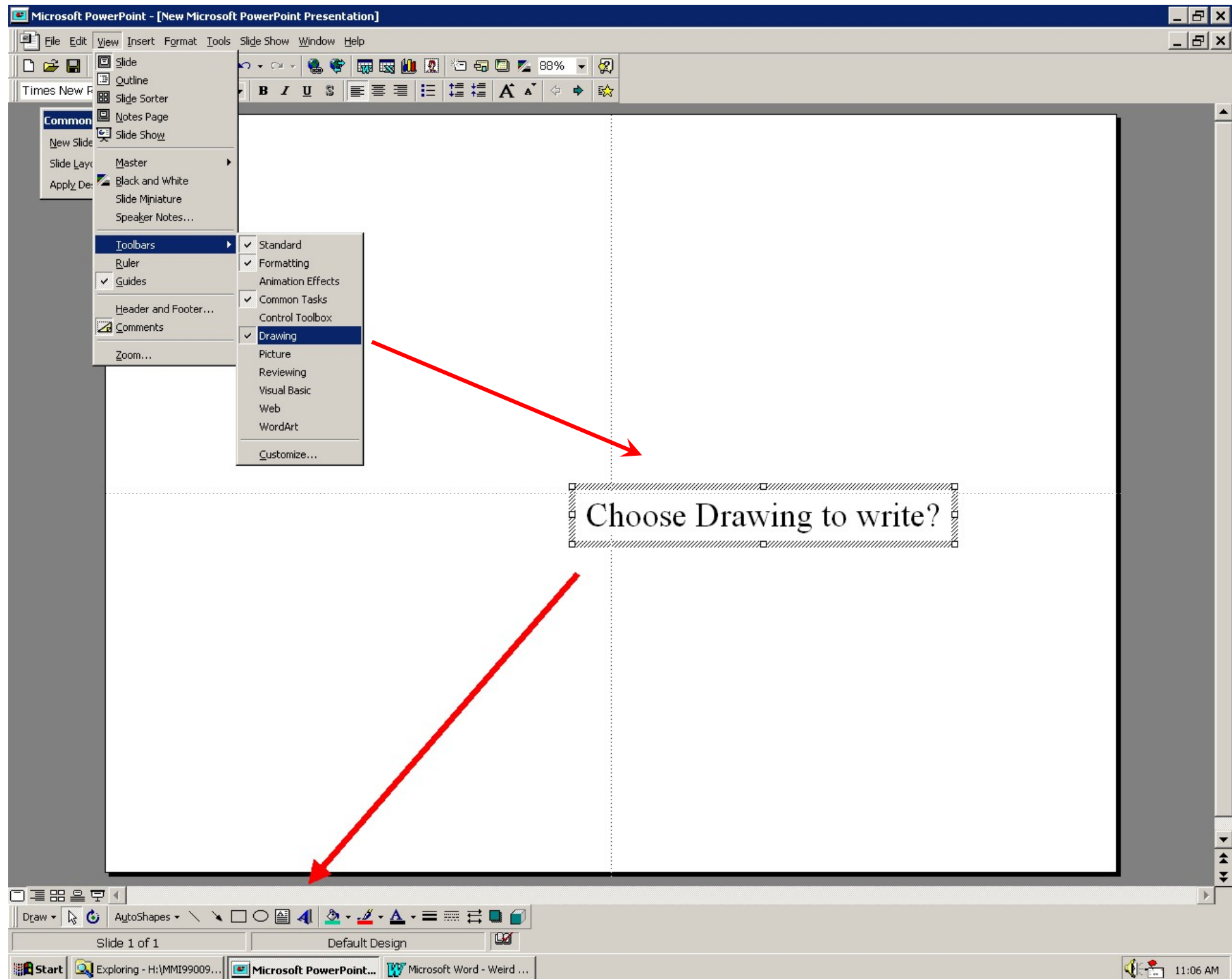
Context!

Context is responsible for the proper semantic prime

Short break:
Next, examples of all kinds of
ambiguity
in text interfaces

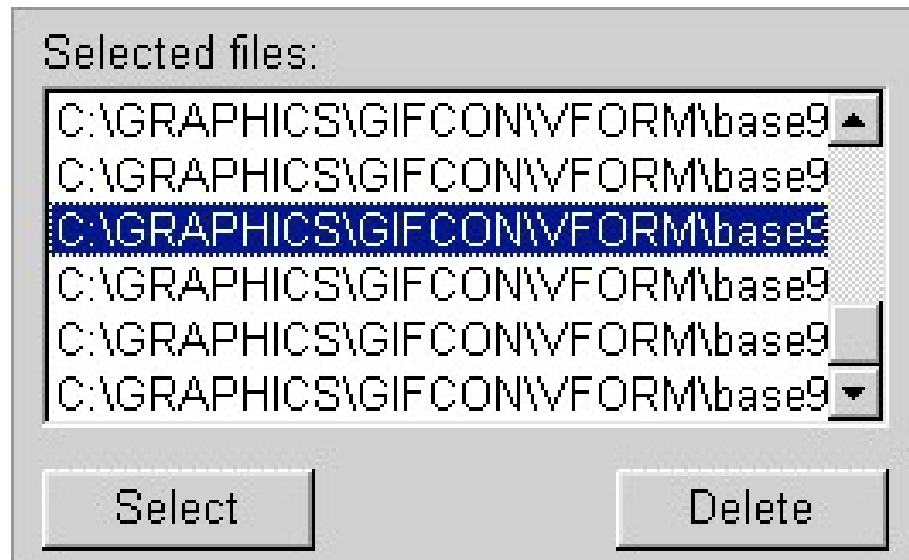


View the non-visual objects: It's magic!

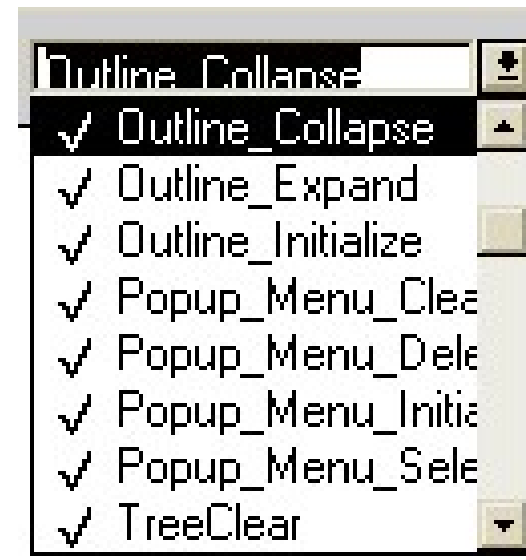


How are you supposed to choose when you can't even discriminate screen elements from each other?

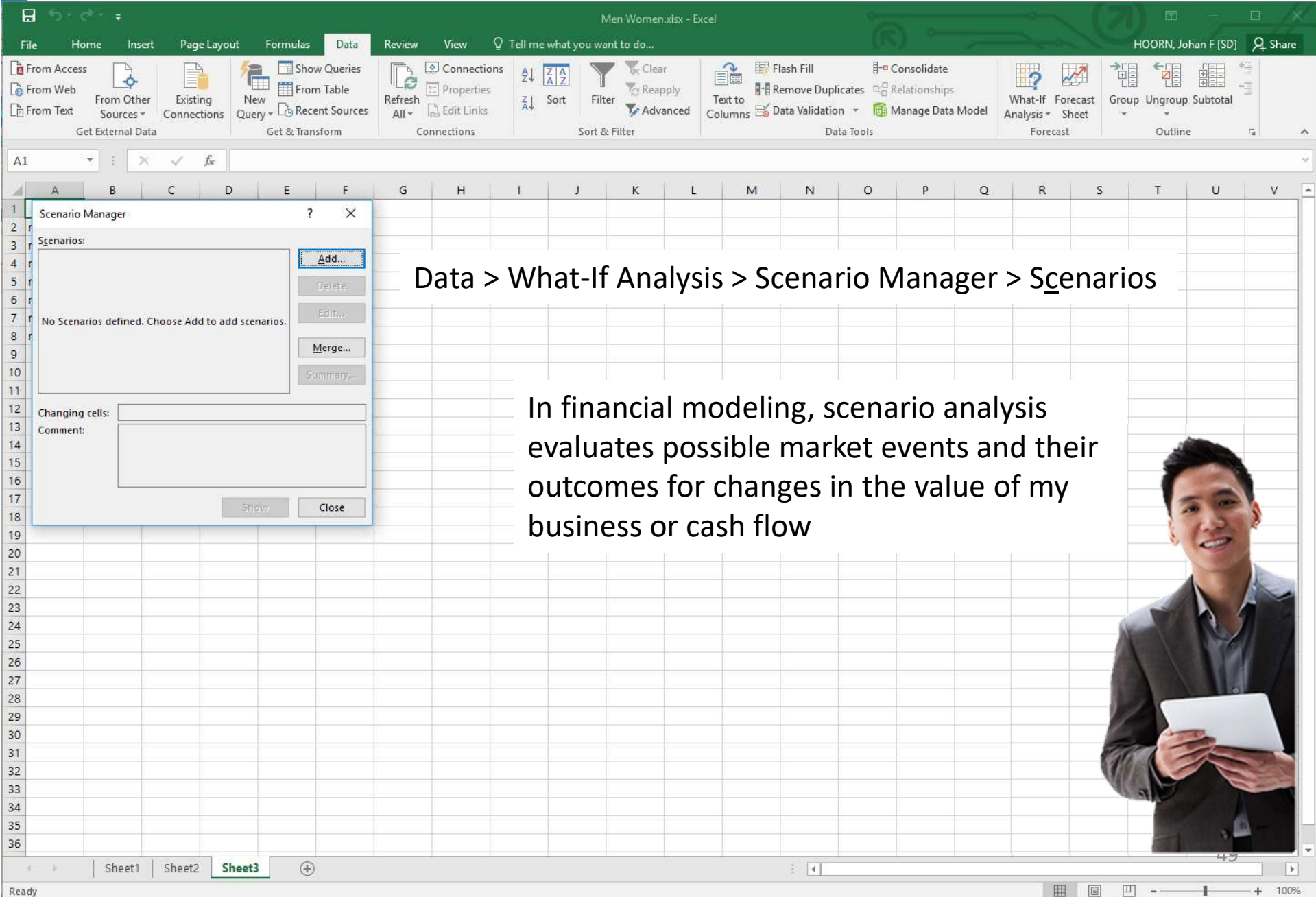
GIF Construction Set



Microsoft Access 2.0



Words mean different things for different people



The screenshot shows the Microsoft Excel interface with the 'Data' tab selected. The 'Scenario Manager' dialog box is open, displaying a list of scenarios (currently empty) and buttons for 'Add...', 'Delete', 'Edit...', 'Merge...', and 'Summary...'. The 'Changing cells' field is empty, and the 'Comment' field is also empty. The 'Show' and 'Close' buttons are at the bottom of the dialog.

Data > What-If Analysis > Scenario Manager > Scenarios

In financial modeling, scenario analysis evaluates possible market events and their outcomes for changes in the value of my business or cash flow



A man in a dark suit and light blue shirt is smiling and holding a white tablet. He is positioned in the bottom right corner of the Excel window.

Words mean different things for different people

Men Women.xlsx - Excel

File Home Insert Page Layout Formulas Data Review View Tell me what you want to do...

From Access From Web From Text From Other Sources Existing Connections New Query Show Queries From Table Recent Sources Get External Data Get & Transform

Connections Refresh All Properties Edit Links Sort Filter Clear Reapply Advanced Text to Columns Flash Fill Remove Duplicates Data Validation Consolidate Relationships Manage Data Model What-If Analysis Forecast Sheet Group Ungroup Subtotal Outline

A1

Scenario Manager

Scenarios:

Add... Delete Edit Merge... Summary...

No Scenarios defined. Choose Add to add scenarios.

Changing cells: Comment:

Show Close

Data > What-If Analysis > Scenario Manager > Scenarios

I use excel sheets for my UX design. Scenarios are tasks, questions, and stories with personas to test the usability of my application against the goals of my users

Visit	Line	Conditions or Directions	Saent	User	Location	Actions	Go to / comments
0	1	"Five minutes of focus" and "Become a member" Challenges have not been completed yet.	Here you can compete and compare yourself with others. But...				
0	2		You first need to complete the "Five minutes of focus", "Become a member", and "Privacy and Data" Challenges. Sorry!				
1	1		Welcome to the Leaderboard (firstname)! Click on someone's name for more details and use the filters at the bottom to view different time periods.				
1	2		Welcome back to the Leaderboard. Once you complete the				
2	2	Display this line until "Use Saent for three days in a row" has been completed.	"Use Saent for three days in a row" challenge, you can also create your own Leaderboards!				
normal	1		[Leaderboard intro variations]				
2	2	Display this line until "Use Saent for three days in a row" has been completed.	"Use Saent for three days in a row" challenge, you can also create your own Leaderboards!				

Sheet1 Sheet2 Sheet3

Ready

I'm done R Main menu

Challenges Main menu Back to previous screen

Go to Challenges Back to main menu <<

Challenges Main menu Back to previous screen

100%



Words mean different things for different people

Men Women.xlsx - Excel

File Home Insert Page Layout Formulas Data Review View Tell me what you want to do...

From Access From Web From Text From Other Sources Existing Connections New Query Show Queries From Table Recent Sources Get External Data Get & Transform

Connections Refresh All Properties Edit Links Sort Filter Clear Reapply Advanced Flash Fill Remove Duplicates Text to Columns Data Validation Consolidate Relationships Manage Data Model What-If Analysis Forecast Group Ungroup Subtotal Outline

A1

Scenario Manager

Scenarios:

Add... Delete Edit... Merge... Summary...

No Scenarios defined. Choose Add to add scenarios.

Changing cells:

Comment:

Show Close

Data > What-If Analysis > Scenario Manager > Scenarios

I am a HK movie director. In martial arts, one must choreograph the action very precisely. Otherwise, accidents will happen

Cast and Crew Call - Call Sheet Template MAC.xls

Episode Title

Production Company Credit

Shoot Date Day x of y

Exec. Producer Janet Smith (xxx) xxx-xxxx
Producer James Smith (xxx) xxx-xxxx
Producer John Doe (xxx) xxx-xxxx
Director Jane Doe (xxx) xxx-xxxx
Line Producer John Smith (xxx) xxx-xxxx
Prod Manager Jane Smith (xxx) xxx-xxxx

Production Office (xxx) xxx-xxxx
Address Line 1
Address Line 2

CALL 9 AM

Sample Note 1: Individual call times may vary. **NO SOCIAL MEDIA**CLOSED SET**
Sample Note 2: Questions? Call Jane Doe (xxx) xxx-xxxx
Sample Note 3: Walkie Assignments (TS)
Sample Note 4: Walkie Units: 5

SCENES	SET AND DESCRIPTION	CHARACTER #	D/N	PAGES	LOCATION/NOTES
SET 1	Description 1				
SET 2	Description 2				
SET 3	Description 3				
SET 4	Description 4				
SET 5	Description 5				
SET 6	Description 6				
SET 7	Description 7				

TOTAL PAGES

Page 1 / Page 2 (Crew) / Legend / +

Page Layout View: Ready

Sum=0

Sheet1 Sheet2 Sheet3

Ready



Ambiguity:
One word pointing at more meanings

Scenario Manager...

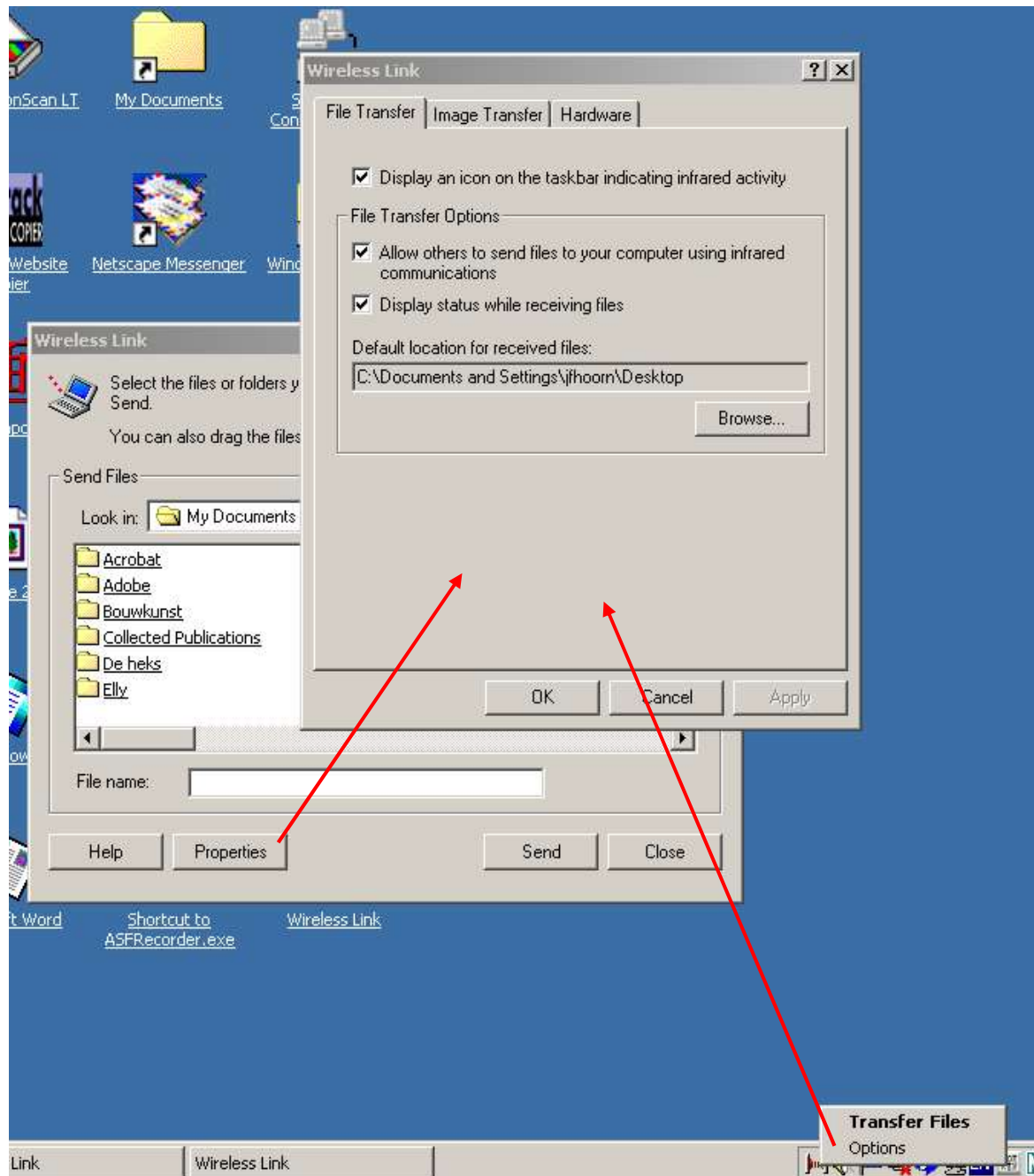
UX designers and
movie directors may
regard themselves as
'scenario managers'

Synonyms:
More words, pointing at one
meaning

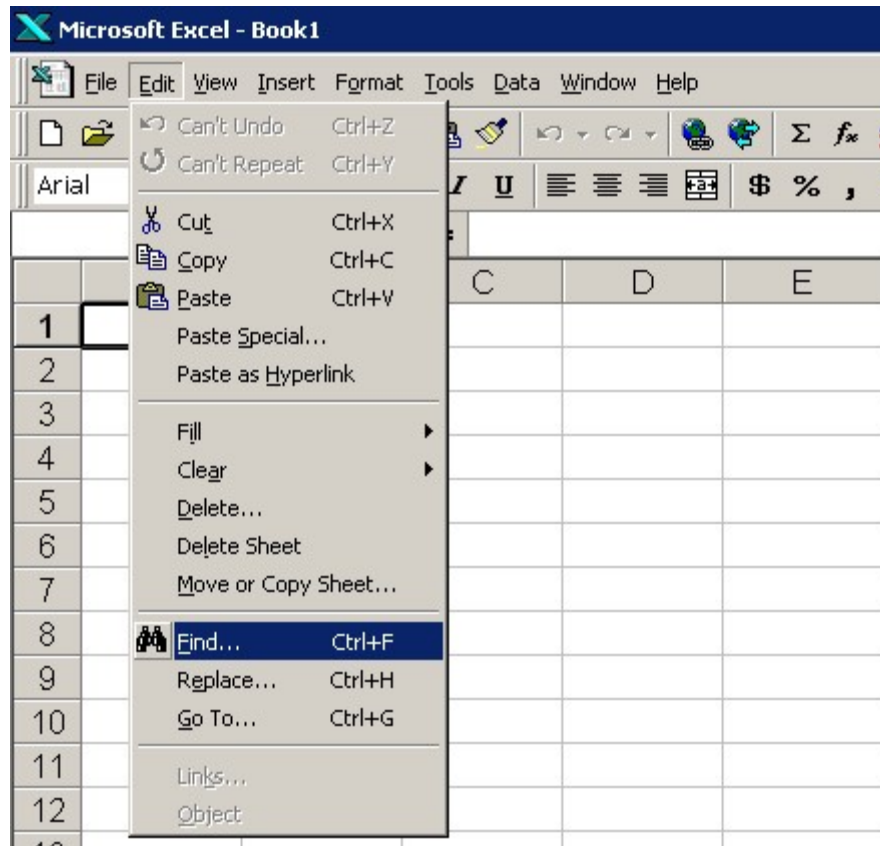
Properties

refer to same window

Options



Random trial-and-error is a low level problem solving strategy but commonly found with non-expert users



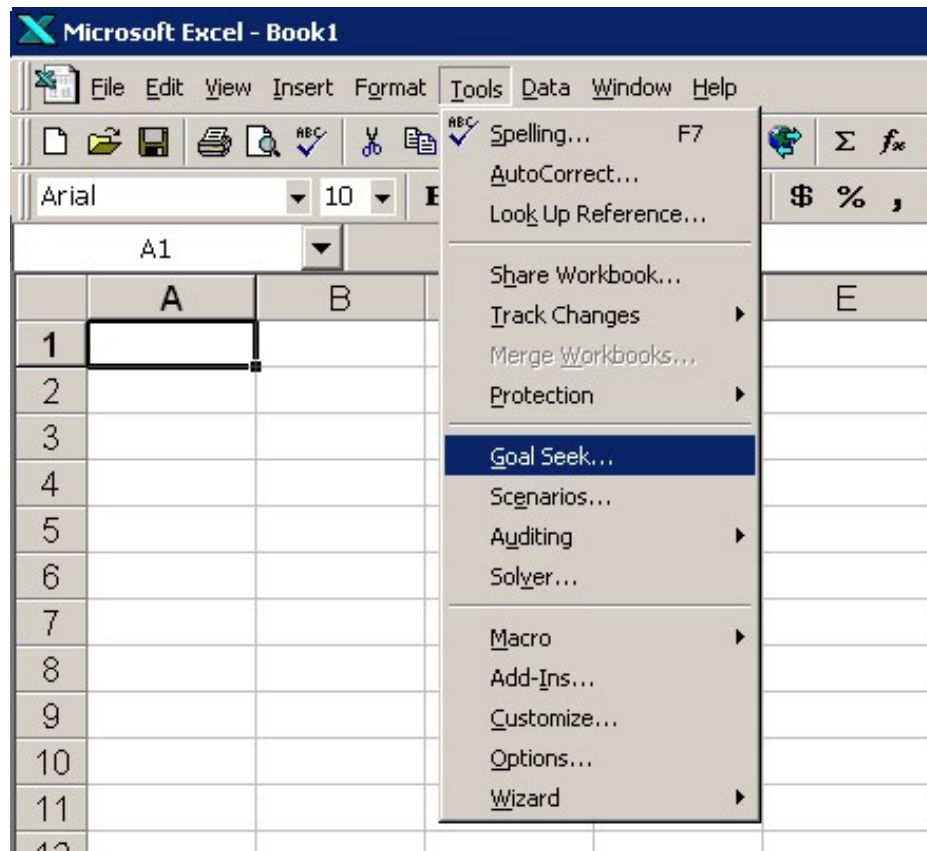
Novices follow first association.

Their problem:

Find (Ctrl F) is not related in meaning to Edit

With the right semantic priming, confused users can be avoided

Random trial-and-error is a low level problem solving strategy but commonly found with non-expert users



Novices follow first association.

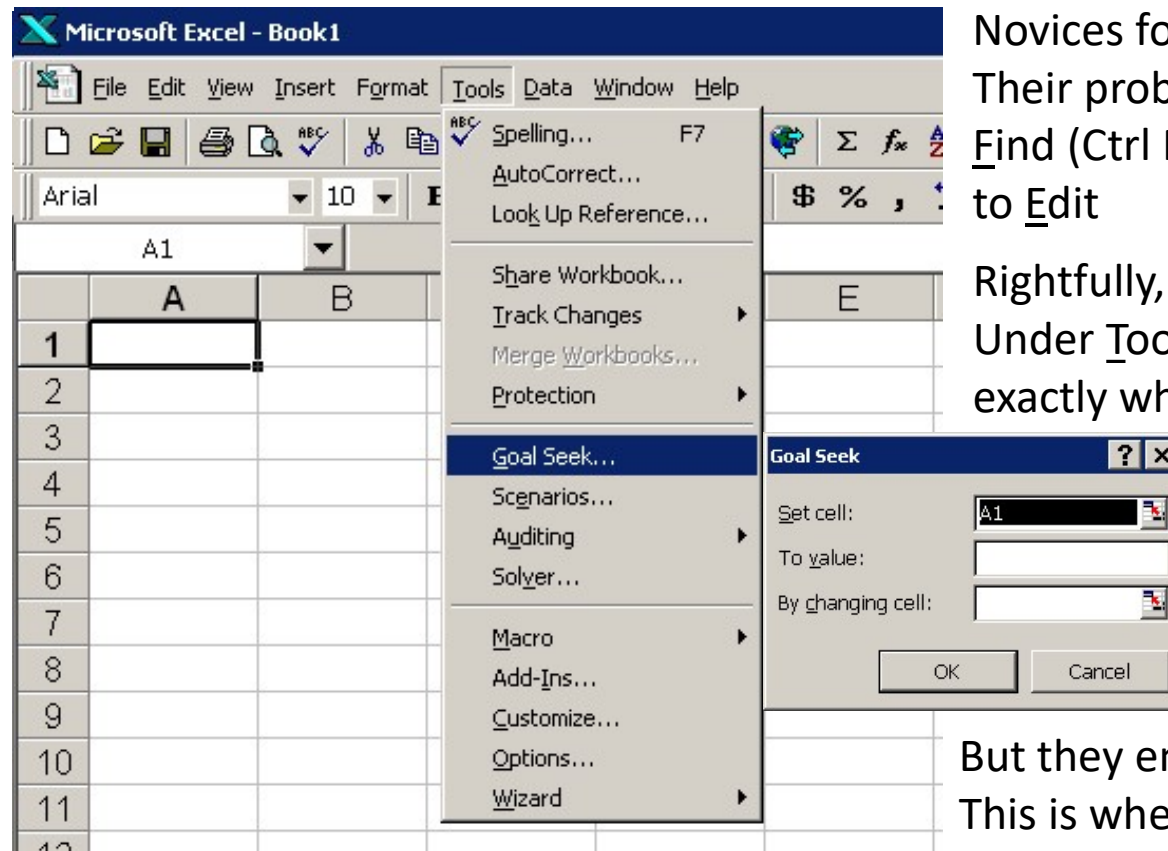
Their problem:

Find (Ctrl F) is not related in meaning to Edit

Rightfully, they see searching as a tool. Under Tools, Goal Seek... describes exactly what they want to do

With the right semantic priming, confused users can be avoided

Random trial-and-error is a low level problem solving strategy but commonly found with non-expert users



Novices follow first association.

Their problem:

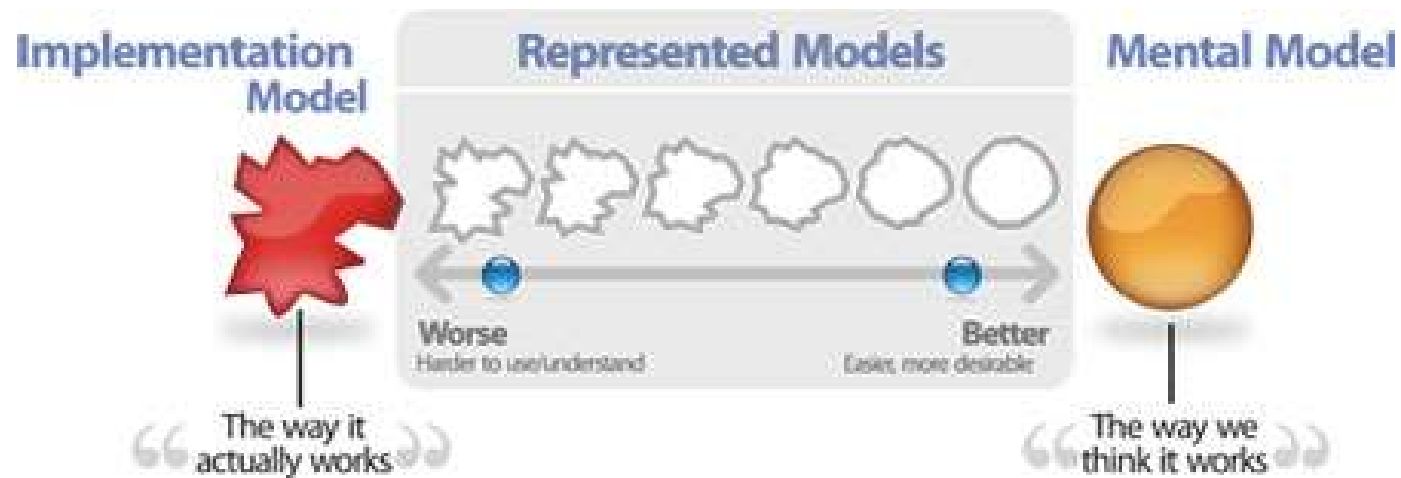
Find (Ctrl F) is not related in meaning to Edit

Rightfully, they see searching as a tool. Under Tools, Goal Seek... describes exactly what they want to do

But they end up setting cells to values
This is where the wild clicking commences...

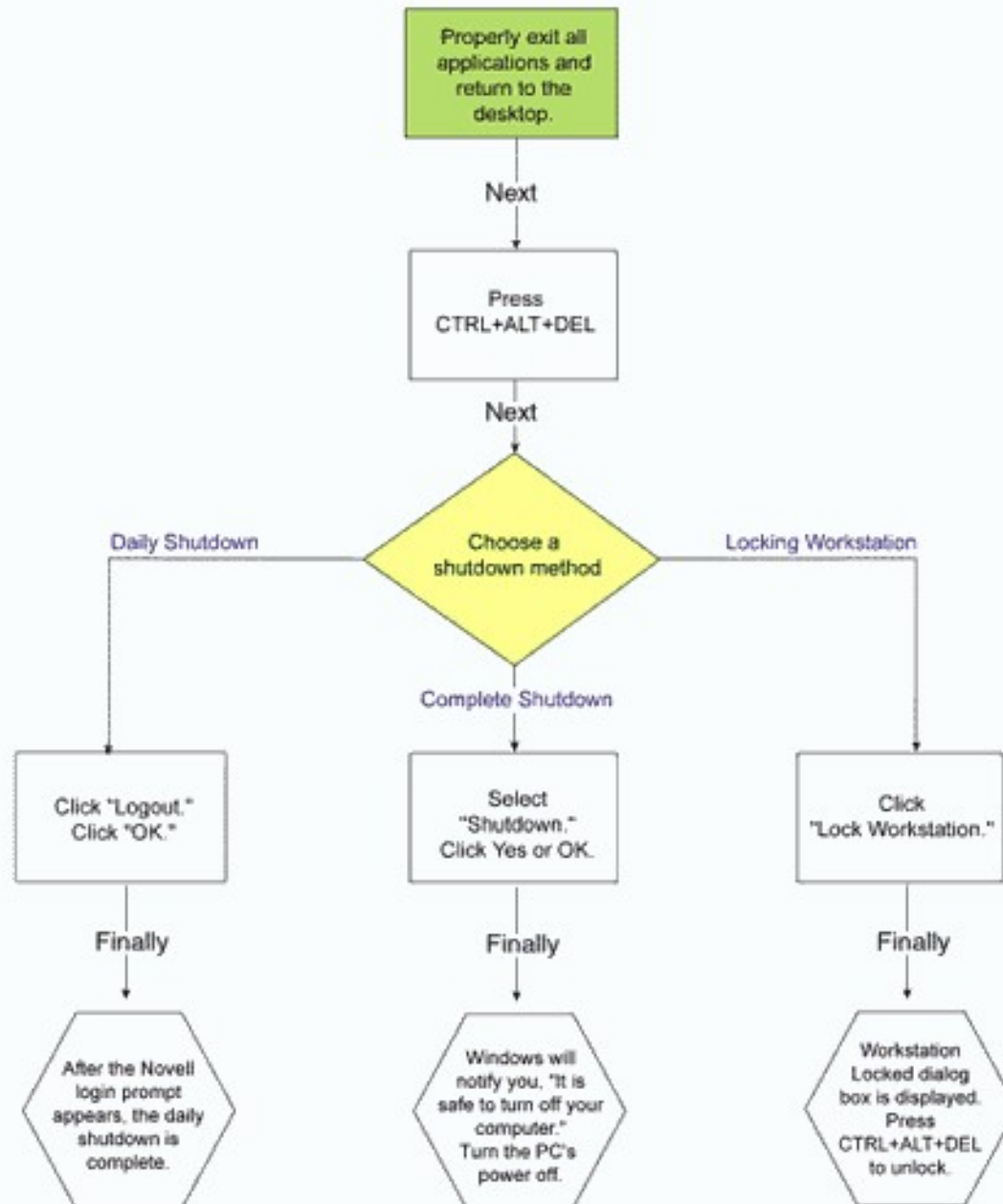
With the right semantic priming, confused users can be avoided

Mental models of designers and users clash

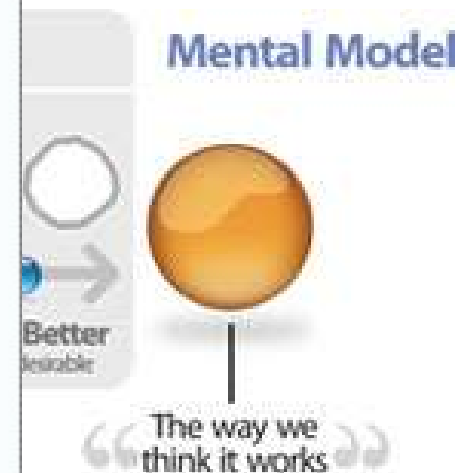


Derived from diagram in *About Face 2.0* (by *Alan Cooper*)

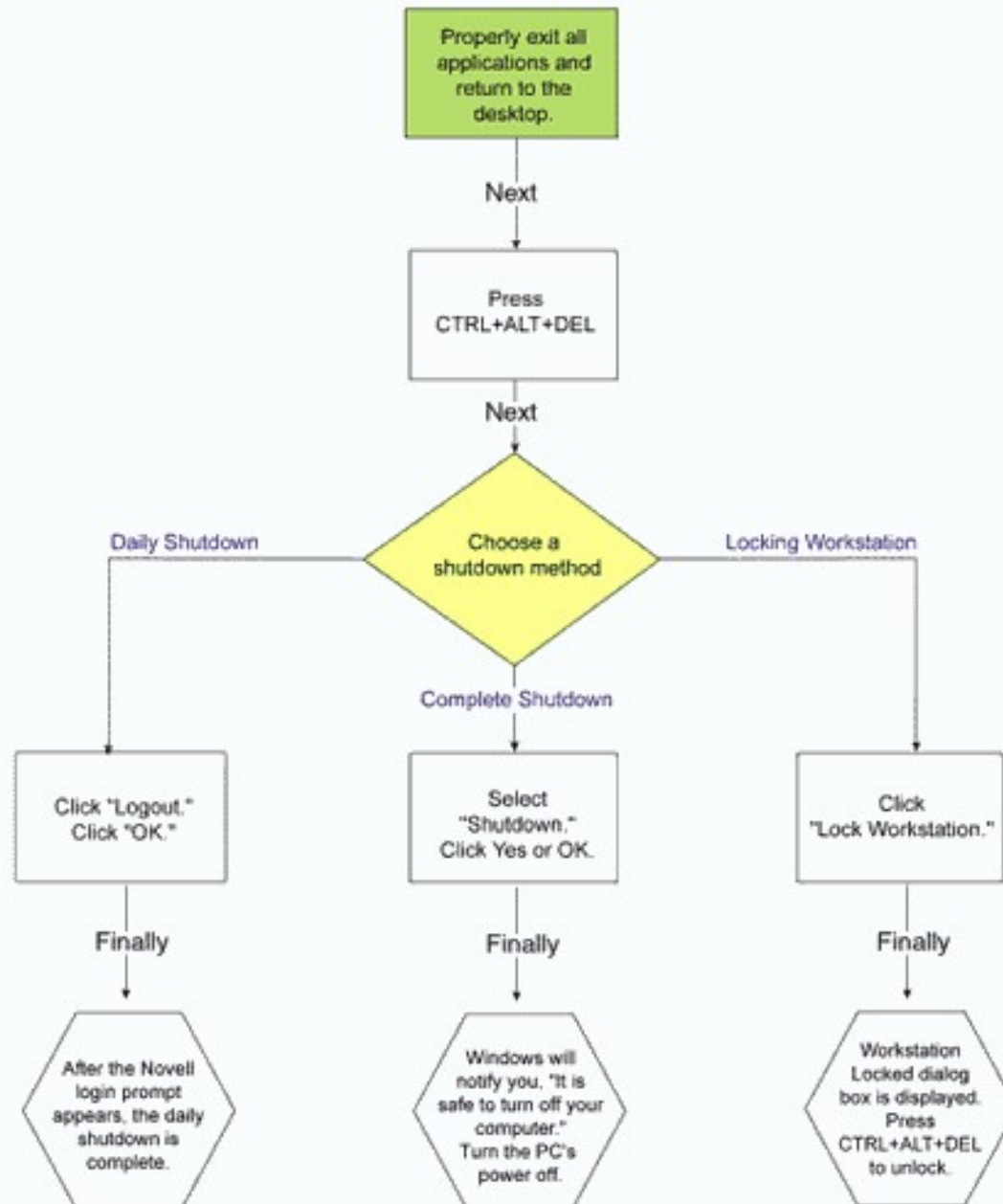
Windows NT 4.0 Shutdown Procedure



rs clash



Windows NT 4.0 Shutdown Procedure



Windows NT 4.0 Shutdown Procedure

Properly exit all applications and return to the desktop.

Next



Daily Shutdown

Click "Logout."
Click "OK."

Finally

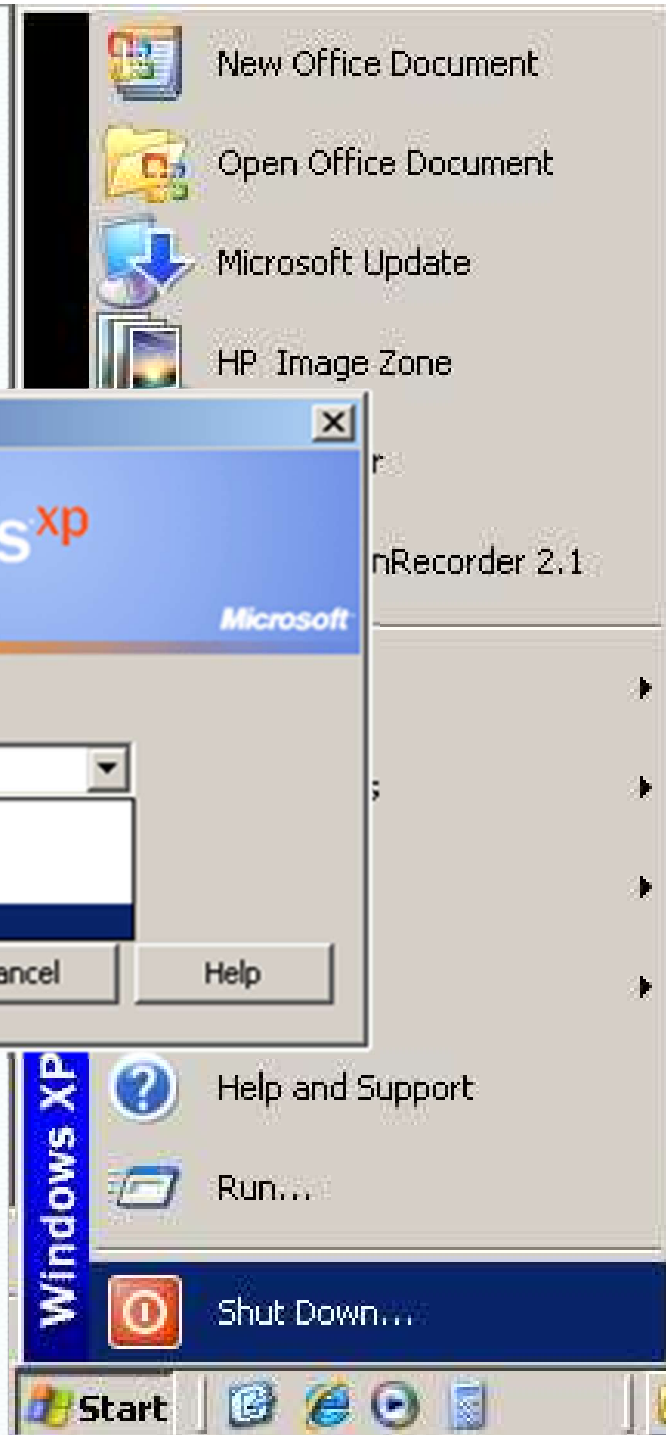
After the Novell login prompt appears, the daily shutdown is complete.

Finally

Windows will notify you. "It is safe to turn off your computer."
Turn the PC's power off.

Finally

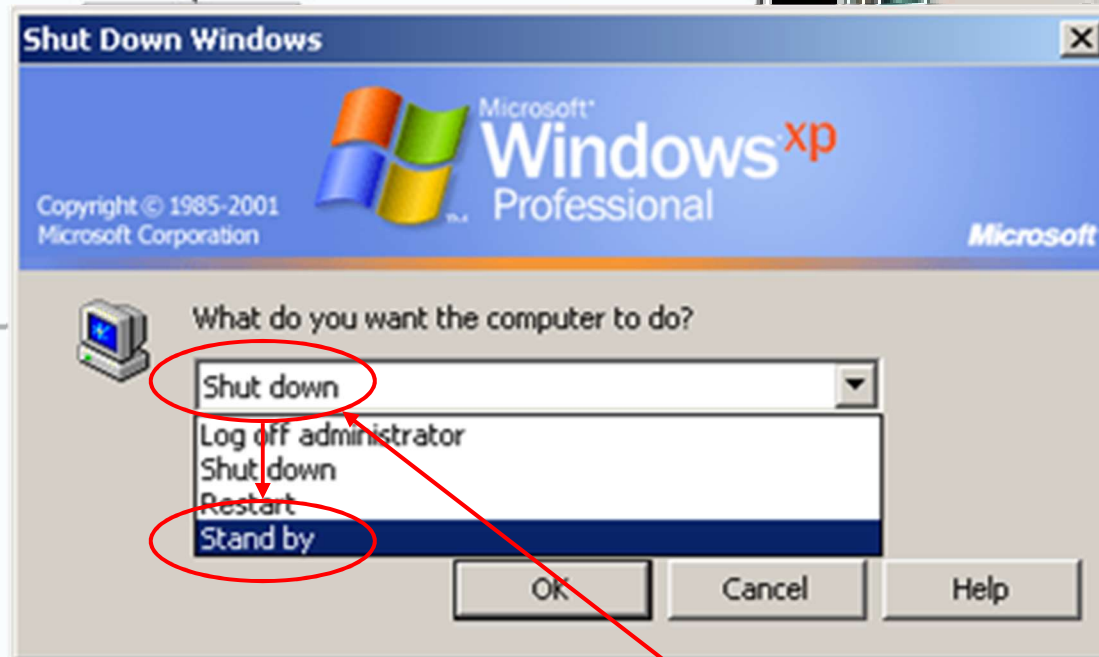
Workstation Locked dialog box is displayed. Press CTRL+ALT+DEL to unlock.



Windows NT 4.0 Shutdown Procedure

Properly exit all applications and return to the desktop.

Next



Daily Shutdown

Click "Logout."
Click "OK."

Finally

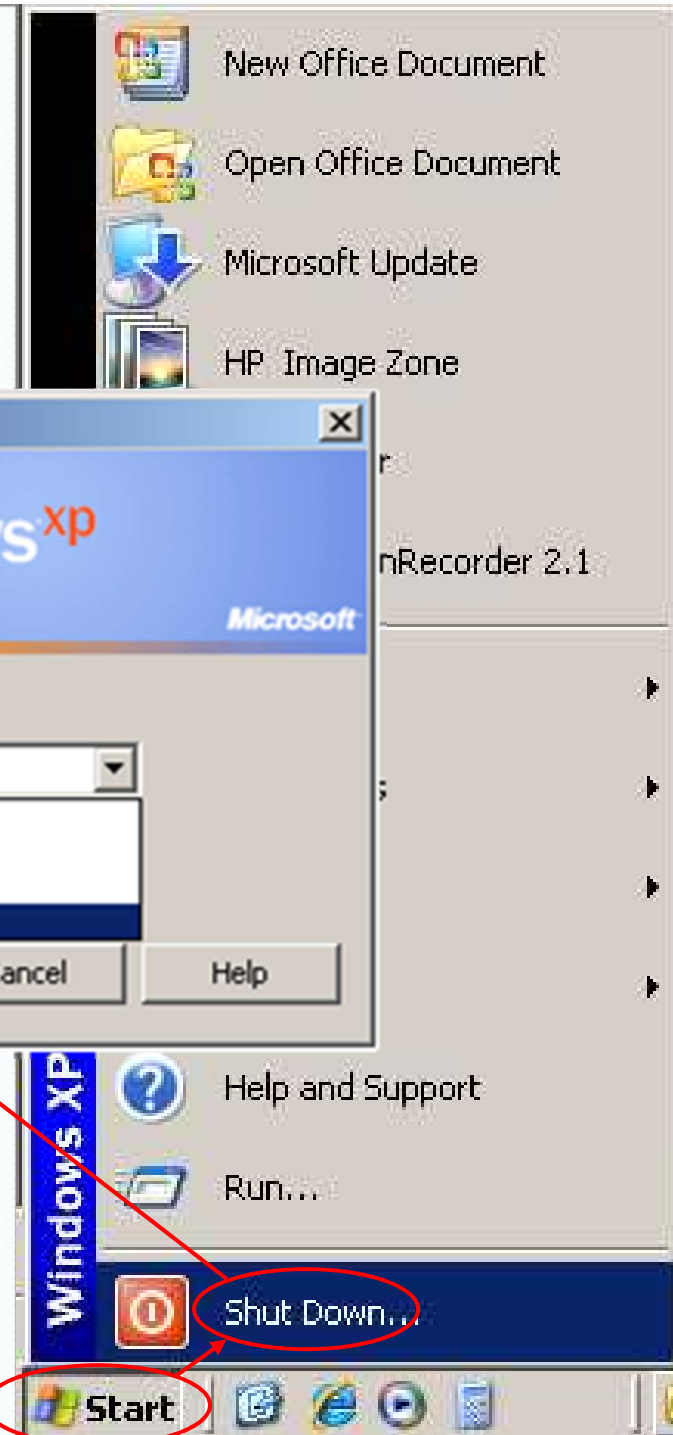
After the Novell login prompt appears, the daily shutdown is complete.

Finally

Windows will notify you. "It is safe to turn off your computer." Turn the PC's power off.

Finally

Workstation Locked dialog box is displayed. Press CTRL+ALT+DEL to unlock.



Blackboard: To Open is to Download?

du.hk/ultra/courses/_109423_1/cl/outline

Download Package Copy Move Delete Refresh Page 1 of 2

☐	FILE TYPE	NAME	EDITED	SIZE	PERMISSIONS
☐	Folder	Subject_e_Eng	Aug 2, 2023 10:02:16 AM	0	
☐	PDF	01 COMP3423	Aug 29, 2023 6:34:53 PM	7.4 MB	
☐	PDF	01 COMP3423	Sep 15, 2023 2:40:36 PM	10.7 MB	
☐	PDF	02 COMP3423	Sep 18, 2023 10:03:41 AM	7.29 MB	
☐	PDF	02 COMP3423	Sep 25, 2023 10:46:37 AM	6.39 MB	
☐	PDF	03 HCI Visuals	Aug 28, 2023 1:39:36 PM	11.1 MB	
☒	PPT	03-COMP3423 VR_2022.pptx	Aug 28, 2023 1:40:12 PM	32.2 MB	

Context menu for the selected file (03-COMP3423 VR_2022.pptx):

- Open
- 360° View
- Edit Settings
- Overwrite File
- Permissions
- Alignments (None)
- Download Package
- Copy
- Move
- Delete

A red arrow points from the 'Open' option in the context menu to the download icon in the top right corner of the interface.

Blackboard: To Open is to Download?

du.hk/ultra/courses/_109423_1/cl/outline

Download Package Copy Move Delete Refresh Page 1 of 2

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Red arrows indicate the flow from the 'Open' option in the context menu to the 'Open' button in the top right corner of the interface.

And how about Proximity of related interface widgets?

Input: perception

□ Visual

- Text
 - Recommendations

Know your user groups (novices, experts, other than intended)

Labels are short

Yet, avoid ambiguity

Don't use synonyms

Labels are self-explicating

Otherwise, labels should prime high-frequency meanings

(Start > Shut Down > Standby > OK???)

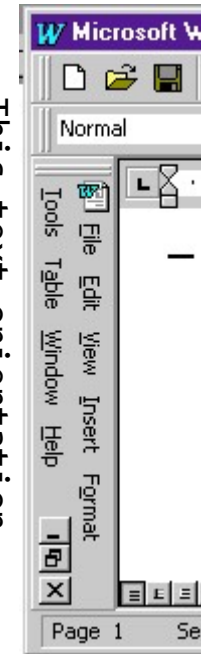
Use context cues (e.g., a text processor activates different associations than a spreadsheet)

- Recommendations

Typography

- Style and appearance of text
- Characters, symbols, graphical elements should be easily noticeable and distinguishable
- Serif fonts are easier to read in print, sans-serif fonts are easier to read on the screen and are more casual
- Proportional fonts take less space and are more legible
- Mixed-case is more compact and easier to read
- Lines should be at most 60 characters long for printed matter
- Don't overdo styles
- Typographic distinction should be used only if it conveys extra information
- Maximum of 1-2 typefaces (3 max), two slants (normal, italic), two weights (normal, bold), four sizes (title, subtitle, text, footnote)
- Distinction between font sizes should be at least 2 pt to be distinguishable to the eye
- Use indentation and color carefully
- Provide sufficient contrast between foreground and background
- Background should be a neutral color (black, white, grey) unless you have a good reason to do otherwise

This text orientation
is difficult to read



CRAP

❑ Contrast

- Make different things different!
- Bring out dominant elements
- Mute lesser elements
- Create dynamism

❑ Repetition

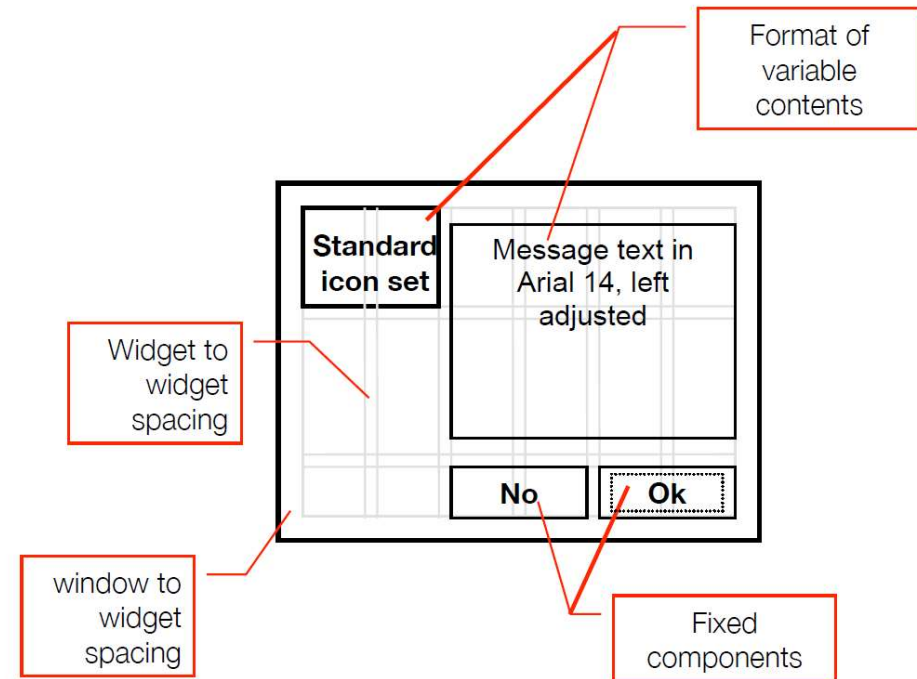
- Repeat design throughout the interface
- Be consistent

❑ Create Unity

- Alignment
- Visually connect elements
- Create visual flow

❑ Proximity

- Group related elements
- Separate unrelated ones



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I only want to log in

Grids

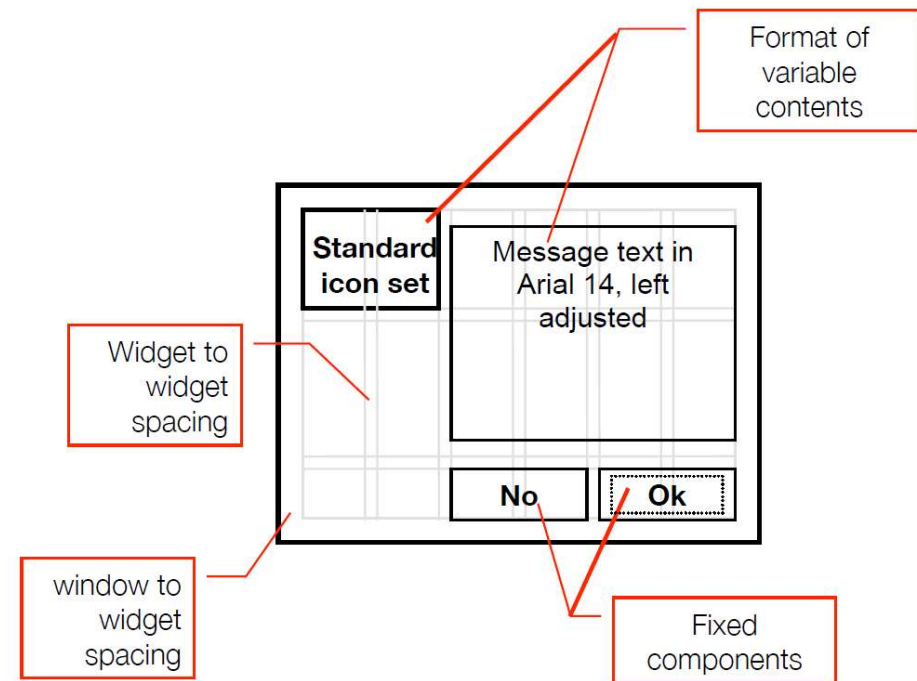
- Horizontal and vertical lines to locate interface components
- Aligns related components

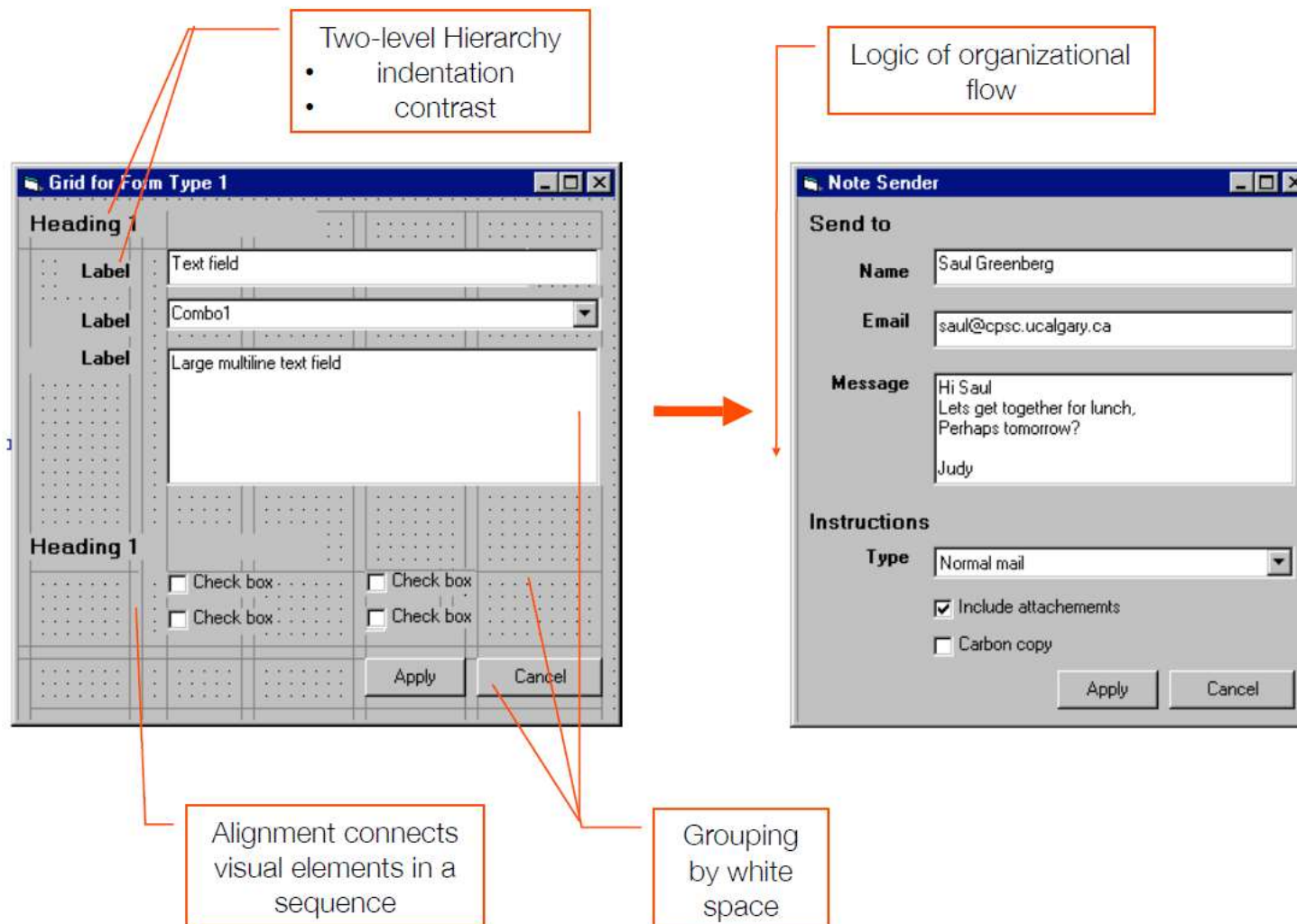
❑ Organization

- Contrast for dominant elements
- Element groupings by proximity
- Organizational structure
- Alignment

❑ Consistency

- Location
- Format
- Element Repetition
- Organization





Example of what NOT to do...

- Terrible alignment!
- No flow
- Poor contrast
- Cannot distinguish colored labels from editable fields
- Poor repetition
- Buttons don't look like buttons!
- Poor explicit structure
- Blocks compete with alignment

Webforms

Form Title -- (appears above URL in most browsers and is used by WWW search)		Background Color:
Q&D Software Development Order Desk		FFFBF0
Form Heading -- (appears at top of Web page in bold type)		Text Color:
Q&D Software Development Order Desk		000080
E-Mail responses to (will not appear on)	Alternate (for mailto forms only)	Background Graphic
dversch@q-d.com		
Text to appear in Submit button	Text to appear in Reset button	☐ Mailto
Send Order	Clear Form	☒ CGI
Scrolling Status Bar Message (max length = 200 characters)		
WebMania 1.5b with Image Map Wizard is here!		
<< Prev Tab		Next Tab >>

- Recommendations

❑ Images: signs, icons, symbols



- Spectrum from concrete to abstract
- Icon design is very hard (go to School of Design)
- Except for the most familiar ones, always label your icons
- Image position and type should be related
- Group images into “families”
- Image use should be consistent and relevant
- Silhouette of the icon conveys the most information
- Design for recognition over recall
- Stick with conventions
- Design first in b/w, add color later



Partial icon family

Virtual Tutor Summary



Information is not neutral. The eye filters, selects, and changes data so to make an interpretation of the world about us. Color is not a property of light but of the eye's receptors. 3D vision results from manipulations by the brain to create a depth illusion. The brain uses heuristics such as similarity, continuity, and proximity to make sense of information that passes the eye. Patterns are not the result of recognition but rather of human construction

Text is a form of visual input, most important to guide the user through the interface (labels to icons included). Text is strongly ambiguous. You should provide elements that semantically prime the intended meaning. This is 'context.' Pressing 'start' to stop a process is priming the wrong semantic associations

Virtual Tutor Summary



To make a GUI, work from:

- ☐ Contrast
- ☐ Repetition
- ☐ Unity
- ☐ Proximity

Grids are a good way to satisfy these requirements

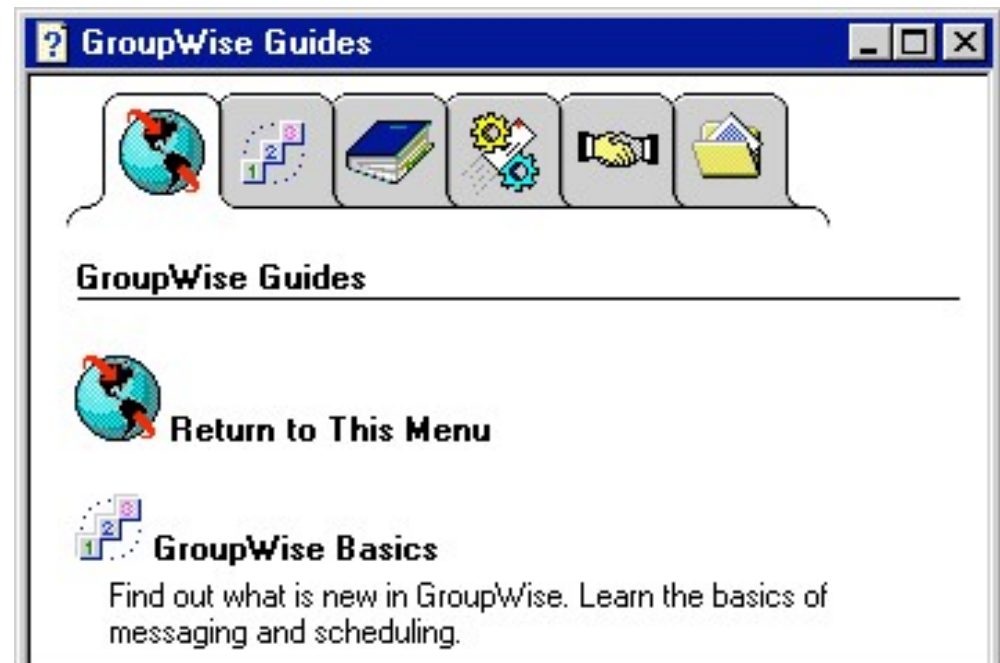
Arrange functions as represented by icons and texts into families so that users can 'chunk up' the information into meaningful units and are not overwhelmed by numerous unrelated widgets

Remember that an image says more than a thousand words but what it says we don't know because words are way more precise than images (that's why humans invented language)

Exam Sample Item

Category “Application”

In Novell Groupwise 5.1, point out the problem with the images and identify the tab that has a glossary explaining these images



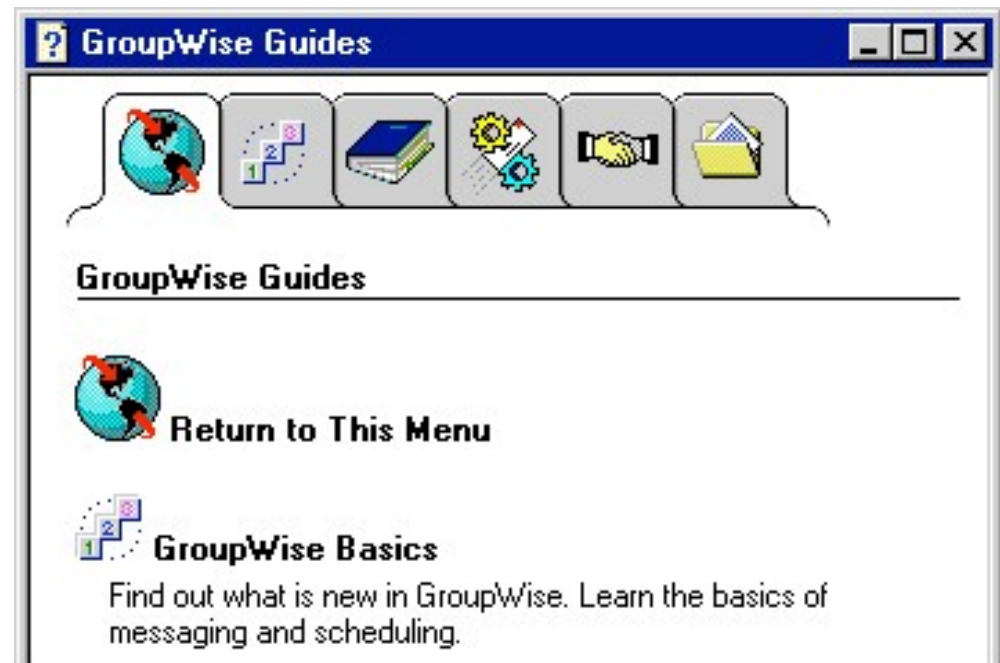
- A What do these images mean? There are no tooltips
- B No element groupings by proximity
- C Items that are similar tend to be interpreted together
- D The images represent ambiguous synonyms

A tooltip tells the function when you hover over

Exam Sample Item

Category “Application”

In Novell Groupwise 5.1, point out the problem with the images and identify the tab that has a glossary explaining these images



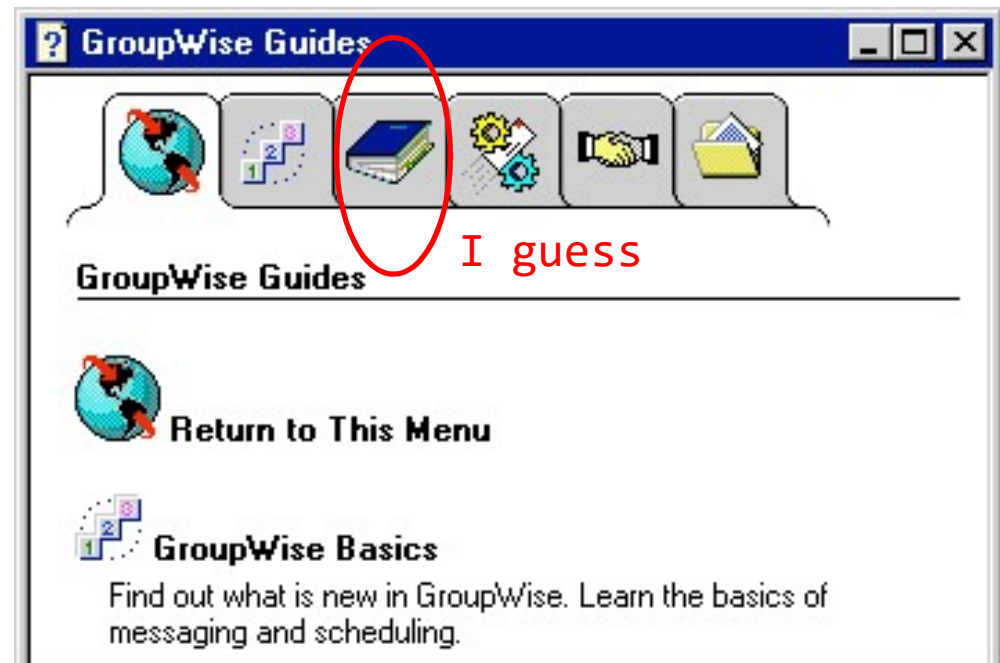
What do these images mean? There are no tooltips

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